Linearly implicit, energy-stable schemes of arbitrary temporal order for a scalar-potential formulation of the Rosensweig model*

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Abstract

We develop linearly implicit, energy-stable schemes of arbitrary temporal order for a scalar-potential formulation of the Rosensweig ferrofluid model. By separating the weak formulation into symmetric dissipative, skew-symmetric bilinear, and skew-symmetric trilinear parts, we extrapolate only the first argument of the nonlinear form and obtain linear Runge–Kutta stage problems without losing the energy cancellation. For algebraically stable Runge–Kutta methods, we prove unique solvability for every positive time step, derive a discrete energy identity, and establish arbitrary-order temporal error estimates under explicit regularity, stage-order, quadrature, and starting-value assumptions. Compatible finite element spaces preserve the energy structure and recover the magnetic field as $\boldsymbol{H}_h = \nabla\varphi_h$, so that its curl vanishes exactly at every stage and time level. Numerical experiments illustrate the convergence and energy decay of the schemes and their application to three-dimensional magnetically driven channel flows.

Keywords: Rosensweig ferrofluid model; linearly implicit Runge–Kutta method; energy stability; high-order time.

1 Introduction

Ferrofluids are liquids containing suspended magnetic particles whose motion can be controlled by an applied magnetic field. This property makes them useful in sealing, damping, heat transfer, and magnetic actuation [21,22]. Two widely used continuum models of ferrohydrodynamics are the Shliomis model [26,27] and the Rosensweig model [21,22]. Both describe ferrofluids as homogeneous single-phase media, but they represent the rotational dynamics of the magnetic particles differently. The Rosensweig model introduces the angular velocity as an independent unknown governed by an angular-momentum equation, whereas the Shliomis model accounts for rotational effects through magnetic-torque terms without introducing a separate angular-velocity equation. This modeling distinction is important when comparing numerical methods and their energy laws.

Because both models consist of strongly coupled nonlinear partial differential equations with nontrivial energy transfer, analytical solutions are rarely available, and numerical approximation is therefore the main practical approach. For the Shliomis model, Wu and Xie [33] developed energy-stable mixed finite element semidiscretizations and fully discrete schemes, proved the existence and uniqueness of the discrete solutions, and derived optimal a priori error estimates. Mao and Sun [15] analyzed a first-order energy-stable semi-implicit Euler finite element method for a simplified Shliomis system, including regularity results and unconditional error estimates. Zhang et al. [37] proposed two linear and unconditionally energy-stable finite element schemes based on hybrid

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continuous–discontinuous and decoupled continuous spatial discretizations, respectively. Keram, Huang, and He [12] further studied a linear, decoupled, variable-step method and established its stability.

For the Rosensweig model, Nochetto, Salgado, and Tomas [18] gave a systematic study of the model and its numerical approximation. They constructed an energy-stable finite element scheme, proved the existence of discrete solutions, and established convergence under additional simplifying assumptions. Wu and Xie [35] developed energy-stable mixed finite element semidiscrete and fully discrete schemes, proved the existence and uniqueness of the discrete solutions, and obtained optimal error estimates based on regularity results. Dong et al. [6] proposed a semi-implicit, energy-stable fully discrete finite element method and proved its unconditional convergence to a weak solution. They later developed a second-order, linearized, and unconditionally energy-stable method whose energy law is written in terms of the original variables [7]. Zhang, He, and Yang [36] introduced a nonlocal auxiliary variable and a zero-energy-contribution decoupling strategy to obtain a fully decoupled, linear, second-order, and unconditionally energy-stable scheme. For a regularized Rosensweig system, Keram, Huang, and He [13] analyzed a linearized, decoupled, variable-step backward Euler method and proved unconditional stability and error estimates.

Beyond the FHD schemes reviewed above, high-order energy-stable Runge–Kutta methods have been developed mainly for gradient-flow problems. Shin et al. [29] combined convex splitting with additive Runge–Kutta methods to construct high-order, unconditionally energy-stable schemes. Akrivis et al. [1] used extrapolation and the scalar auxiliary variable approach to obtain linear Runge–Kutta schemes of arbitrary order, while Shin and Lee [30] developed high-order energy-stable Runge–Kutta methods for convex gradient problems. Gong et al. [8] combined energy quadratization with algebraically stable Runge–Kutta methods to construct arbitrary-order linear schemes. Liu and Yin [28] extended this approach to Runge–Kutta discontinuous Galerkin schemes for fourth-order gradient flows, and Tan and Tang [31] studied a broader class of linear energy-stable schemes based on scalar auxiliary variables and general linear time discretizations. Sato et al. [23] also developed high-order linearly implicit schemes that preserve quadratic invariants. These studies provide useful tools for high-order energy-stable time integration. However, they do not directly cover the Rosensweig system, where the pressure constraint, skew magnetic couplings, curl–div spaces, and scalar-potential boundary conditions must be treated together.

Motivated by the studies above, we develop high-order time discretizations for the Rosensweig model that preserve its energy structure and require only linear stage solves. Introducing the scalar magnetic potential $\boldsymbol{H} = \nabla\varphi$, we write the Rosensweig FHD model in the abstract form

$$\begin{aligned} \mathtt{D}(\partial_t \boldsymbol{x}, \boldsymbol{y}) + \mathtt{A}(\boldsymbol{x}, \boldsymbol{y}) + \mathtt{J}(\boldsymbol{x}, \boldsymbol{y}) + \mathtt{B}(\boldsymbol{x}, \boldsymbol{x}, \boldsymbol{y}) + \mathtt{C}(\boldsymbol{y}, \widetilde{p}) &= \langle \mathtt{f}, \boldsymbol{y} \rangle \,, \\ \mathtt{C}(\boldsymbol{x}, q) &= 0. \end{aligned} \tag{1.1}$$

Here, $\mathtt{D}$ is the mass form, $\mathtt{A}$ is symmetric and coercive, $\mathtt{J}$ is a skew-symmetric bilinear form, $\mathtt{B}$ is trilinear and skew-symmetric in its last two arguments, and $\mathtt{C}$ represents the incompressibility constraint. In particular,

$$\mathtt{J}(\boldsymbol{y}, \boldsymbol{y}) = 0, \qquad \mathtt{B}(\boldsymbol{z}, \boldsymbol{y}, \boldsymbol{y}) = 0.$$

These identities allow us to extrapolate the first argument of $\mathtt{B}$ while treating the current stage state implicitly. Each Runge–Kutta step is therefore linear, and the energy cancellation is preserved.

Based on this structure, we construct linearly implicit Runge–Kutta schemes of arbitrary temporal order. For algebraically stable Runge–Kutta coefficients, we prove unique solvability for every positive time step and derive an exact discrete energy identity. Under the stage and quadrature conditions $C(p-1)$ and $B(p)$, together with sufficient regularity and accurate starting values, we establish order-p temporal error estimates directly in the continuous function spaces. Backward Euler, extrapolated Crank–Nicolson, and third- and fourth-order Gauss-type methods are obtained as special cases. We also develop compatible finite element schemes using MINI or Taylor–Hood elements for velocity and pressure, and Nédélec and Raviart–Thomas elements for the magnetic variables. The fully discrete schemes retain the energy estimate without a restriction between the

time step and the mesh size. Moreover,

$$\boldsymbol{H}_h^n = \nabla\varphi_h^n, \qquad \boldsymbol{H}_{h,n,i} = \nabla\varphi_{h,n,i},$$

so that

$$\operatorname{curl}\boldsymbol{H}_h^n = 0, \qquad \operatorname{curl}\boldsymbol{H}_{h,n,i} = 0$$

at every time level and Runge–Kutta stage. This exact curl-free property does not, by itself, enforce the magnetostatic divergence relation.

The remainder of the paper is organized as follows. Section 2 introduces the Rosensweig model, function spaces, and scalar-potential boundary closure. Section 3 defines the abstract variational forms and their structural properties. Section 4 develops the arbitrary-order time discretization and presents the first- through fourth-order examples. Section 5 gives the compatible finite element schemes, their implementation, and the block elimination. Section 6 presents the convergence, energy, and three-dimensional flow experiments.

2 Model, spaces, and boundary compatibility

2.1 The Rosensweig equations

Let $\Omega \subset \mathbb{R}^3$ be a bounded, simply connected, and convex domain with Lipschitz boundary $\partial\Omega$ and outward unit normal $\boldsymbol{n}$. Let $T > 0$ be the final time. We consider an incompressible, viscous Newtonian ferrofluid occupying the domain Ω and subject to an externally applied magnetic field $\boldsymbol{H}_e$ satisfying the boundary condition $\boldsymbol{H}_e \cdot \boldsymbol{n} = 0$ on $\partial\Omega \times (0,T]$. The unknowns are the velocity $\boldsymbol{u}$, pressure p, angular velocity $\boldsymbol{\omega}$, magnetization $\boldsymbol{m}$, and magnetic field $\boldsymbol{H}$. In $\Omega \times (0,T)$ they satisfy

$$\rho(\partial_t\boldsymbol{u} + (\boldsymbol{u}\cdot\nabla)\boldsymbol{u}) - (\eta+\zeta)\Delta\boldsymbol{u} + \nabla p = \mu_0(\boldsymbol{m}\cdot\nabla)\boldsymbol{H} + 2\zeta\operatorname{curl}\boldsymbol{\omega}, \tag{2.1a}$$

$$\operatorname{div}\boldsymbol{u} = 0, \tag{2.1b}$$

$$\rho\kappa(\partial_t\boldsymbol{\omega} + (\boldsymbol{u}\cdot\nabla)\boldsymbol{\omega}) - \eta'\Delta\boldsymbol{\omega} - (\eta'+\lambda')\nabla\operatorname{div}\boldsymbol{\omega} = \mu_0\boldsymbol{m}\times\boldsymbol{H} + 2\zeta(\operatorname{curl}\boldsymbol{u} - 2\boldsymbol{\omega}), \tag{2.1c}$$

$$\partial_t\boldsymbol{m} + (\boldsymbol{u}\cdot\nabla)\boldsymbol{m} - \sigma\Delta\boldsymbol{m} = \boldsymbol{\omega}\times\boldsymbol{m} - \tau^{-1}(\boldsymbol{m} - \chi_0\boldsymbol{H}), \tag{2.1d}$$

$$\operatorname{curl}\boldsymbol{H} = 0, \tag{2.1e}$$

$$\mu_0\operatorname{div}(\boldsymbol{H}+\boldsymbol{m}) = -\operatorname{div}\boldsymbol{H}_e. \tag{2.1f}$$

All parameters $\rho, \kappa, \eta, \zeta, \eta', \lambda', \mu_0, \sigma, \tau, \chi_0$ are positive. In particular, $\sigma > 0$ is needed for the coercivity used below. With the convention $\boldsymbol{B} = \mu_0(\boldsymbol{H}+\boldsymbol{m})$, equation (2.1f) states that $\boldsymbol{B}+\boldsymbol{H}_e$, rather than $\boldsymbol{B}$ alone, is divergence-free. We prescribe initial values

$$\boldsymbol{u}(0) = \boldsymbol{u}_0, \qquad \boldsymbol{\omega}(0) = \boldsymbol{\omega}_0, \qquad \boldsymbol{m}(0) = \boldsymbol{m}_0, \tag{2.2}$$

with $\operatorname{div}\boldsymbol{u}_0 = 0$, and boundary conditions

$$\boldsymbol{u} = \boldsymbol{\omega} = 0, \qquad \boldsymbol{m}\cdot\boldsymbol{n} = 0, \qquad (\operatorname{curl}\boldsymbol{m})\times\boldsymbol{n} = 0, \qquad \boldsymbol{H}\times\boldsymbol{n} = 0 \quad \text{on } \partial\Omega. \tag{2.3}$$

The condition on $\operatorname{curl}\boldsymbol{m}$ is the natural boundary condition for the curl–div realization of the magnetization diffusion.

2.2 Function spaces and norms

We write $(\cdot,\cdot)$ for the scalar or vector $L^2(\Omega)$ inner product. Standard Sobolev norms are denoted by $\|\cdot\|_{H^s}$; a norm without a subscript is the L^2 norm. Set

$$\boldsymbol{S} = H_0^1(\Omega)^3, \qquad \Sigma = H_0^1(\Omega), \qquad W = L_0^2(\Omega),$$
$$\boldsymbol{U} = H_0(\operatorname{curl};\Omega), \qquad \boldsymbol{V} = H_0(\operatorname{div};\Omega),$$
$$D(-\Delta_D) = \{\psi\in\Sigma : \Delta\psi \in L^2(\Omega)\},$$
$$\mathbb{V} = \boldsymbol{S}\times\boldsymbol{S}\times(\boldsymbol{V}\cap H(\operatorname{curl};\Omega))\times D(-\Delta_D).$$

Here $H_0(\mathrm{curl})$ imposes a zero tangential trace, $H_0(\mathrm{div})$ imposes a zero normal trace, and L_0^2 imposes zero mean.

For $\boldsymbol{y} = (\boldsymbol{v}, \boldsymbol{s}, \boldsymbol{F}, \psi) \in \mathbb{V}$, let

$$\|\boldsymbol{y}\|_{\mathbb{V}}^2 = \|\nabla \boldsymbol{v}\|^2 + \|\nabla \boldsymbol{s}\|^2 + \|\boldsymbol{F}\|^2 + \|\operatorname{div} \boldsymbol{F}\|^2 + \|\operatorname{curl} \boldsymbol{F}\|^2 + \|\nabla \psi\|^2 + \|\Delta \psi\|^2. \tag{2.4}$$

We assume that Ω is sufficiently smooth so that, for all $\boldsymbol{F} \in (H_0(\mathrm{curl}) \cap H(\mathrm{div})) \cup (H(\mathrm{curl}) \cap H_0(\mathrm{div}))$ and $\psi \in H_0^1(\Omega) \cap H^2(\Omega)$, the following spatial regularity estimates hold:

$$\|\boldsymbol{F}\|_{H^1} \le C(\|\operatorname{curl} \boldsymbol{F}\| + \|\operatorname{div} \boldsymbol{F}\|), \qquad \|\psi\|_{H^2} \le C\|\Delta \psi\|. \tag{2.5}$$

2.3 Scalar potential and the additional boundary closure

Equation (2.1e), together with the boundary condition $\boldsymbol{H} \times \boldsymbol{n} = 0$ on $\partial\Omega \times (0, T]$ and Equation (2.1f), implies there exists a unique scalar magnetic potential $\varphi \in H_0^1(\Omega)$ such that

$$\boldsymbol{H} = \nabla \varphi, \qquad \mu_0 \Delta \varphi + \mu_0 \operatorname{div} \boldsymbol{m} = -\operatorname{div} \boldsymbol{H}_e. \tag{2.6}$$

The initial value of the potential φ is fixed by

$$-\mu_0 \Delta \varphi_0 = \mu_0 \operatorname{div} \boldsymbol{m}_0 + \operatorname{div} \boldsymbol{H}_e(0), \qquad \varphi_0|_{\partial\Omega} = 0. \tag{2.7}$$

Define

$$\alpha = \frac{\mu_0(1 + \chi_0) + \chi_0}{\tau}, \qquad \beta = \frac{\mu_0 + \chi_0}{\mu_0 \tau}. \tag{2.8}$$

Differentiating the last equation of (2.6), using (2.1d), and adding a multiple of the same constraint gives the bulk equation

$$\begin{aligned} &-\mu_0 \Delta \partial_t \varphi - \mu_0 \operatorname{div}(\boldsymbol{\omega} \times \boldsymbol{m}) - \frac{\chi_0}{\tau} \operatorname{div} \boldsymbol{m} - \alpha \Delta \varphi \\ &\quad + \mu_0 \operatorname{div}\big((\boldsymbol{u} \cdot \nabla)\boldsymbol{m}\big) + \mu_0 \sigma \Delta^2 \varphi = \operatorname{div} \partial_t \boldsymbol{H}_e + \beta \operatorname{div} \boldsymbol{H}_e - \sigma \Delta \operatorname{div} \boldsymbol{H}_e. \end{aligned} \tag{2.9}$$

Equation (2.9) is fourth order in space with respect to φ, the existing boundary condition $\varphi = 0$ on $\partial\Omega$ alone is insufficient to specify the boundary value problem. We therefore impose an artificial boundary condition on $\Delta\varphi$, i.e., we consider the Navier boundary condition for φ:

$$\varphi = 0, \qquad \Delta \varphi = 0 \quad \text{on } \partial\Omega. \tag{2.10}$$

3 An abstract formulation

3.1 The dissipative and skew-symmetric forms

Let

$$\boldsymbol{x} = (\boldsymbol{u}, \boldsymbol{\omega}, \boldsymbol{m}, \varphi), \quad \boldsymbol{y} = (\boldsymbol{v}, \boldsymbol{s}, \boldsymbol{F}, \psi), \quad \boldsymbol{z} = (\boldsymbol{\epsilon}, \boldsymbol{\xi}, \boldsymbol{A}, \phi).$$

We use the scalar transport forms

$$b(\boldsymbol{\epsilon}, \boldsymbol{u}, \boldsymbol{v}) = \frac{1}{2}\big(((\boldsymbol{\epsilon} \cdot \nabla)\boldsymbol{u}, \boldsymbol{v}) - ((\boldsymbol{\epsilon} \cdot \nabla)\boldsymbol{v}, \boldsymbol{u})\big), \tag{3.1}$$

$$c(\boldsymbol{\epsilon}, \boldsymbol{m}, \boldsymbol{F}) = \frac{1}{2}\big((\boldsymbol{\epsilon} \cdot \boldsymbol{m}, \operatorname{div} \boldsymbol{F}) - (\boldsymbol{\epsilon} \cdot \boldsymbol{F}, \operatorname{div} \boldsymbol{m})\big). \tag{3.2}$$

The mass and dissipative forms are

$$\mathtt{D}(\boldsymbol{x}, \boldsymbol{y}) = \rho(\boldsymbol{u}, \boldsymbol{v}) + \rho\kappa(\boldsymbol{\omega}, \boldsymbol{s}) + (\boldsymbol{m}, \boldsymbol{F}) + \mu_0(\nabla \varphi, \nabla \psi) \tag{3.3}$$

$$\begin{aligned} \mathtt{A}(\boldsymbol{x}, \boldsymbol{y}) = {} & \eta(\nabla \boldsymbol{u}, \nabla \boldsymbol{v}) + \eta'(\nabla \boldsymbol{\omega}, \nabla \boldsymbol{s}) + (\eta' + \lambda')(\operatorname{div} \boldsymbol{\omega}, \operatorname{div} \boldsymbol{s}) \\ & + \sigma(\operatorname{curl} \boldsymbol{m}, \operatorname{curl} \boldsymbol{F}) + \sigma(\operatorname{div} \boldsymbol{m}, \operatorname{div} \boldsymbol{F}) + \tau^{-1}(\boldsymbol{m}, \boldsymbol{F}) \\ & + \alpha(\nabla \varphi, \nabla \psi) + \sigma\mu_0(\Delta \varphi, \Delta \psi) + \zeta(\operatorname{curl} \boldsymbol{u} - 2\boldsymbol{\omega}, \operatorname{curl} \boldsymbol{v} - 2\boldsymbol{s}). \end{aligned} \tag{3.4}$$

We separate the linear skew coupling and the pressure constraint:

$$\mathtt{J}(\boldsymbol{x},\boldsymbol{y}) = \frac{\chi_0}{\tau}\big((\boldsymbol{m},\nabla\psi) - (\boldsymbol{F},\nabla\varphi)\big), \qquad \mathtt{C}(\boldsymbol{y},q) = -(\operatorname{div}\boldsymbol{v},q). \tag{3.5}$$

The nonlinear form is genuinely trilinear:

$$\begin{aligned}\mathtt{B}(\boldsymbol{z},\boldsymbol{x},\boldsymbol{y}) = {}& \rho b(\boldsymbol{\epsilon},\boldsymbol{u},\boldsymbol{v}) + \rho\kappa b(\boldsymbol{\epsilon},\boldsymbol{\omega},\boldsymbol{s}) - c(\boldsymbol{\epsilon},\boldsymbol{m},\boldsymbol{F}) - \tfrac{1}{2}(\boldsymbol{m}\times\operatorname{curl}\boldsymbol{\epsilon},\boldsymbol{F}) - (\boldsymbol{\xi}\times\boldsymbol{m},\boldsymbol{F})\\ &+ \tfrac{\mu_0}{2}\big\{(\nabla\varphi,\boldsymbol{v}\operatorname{div}\boldsymbol{A}) - (\nabla\psi,\boldsymbol{u}\operatorname{div}\boldsymbol{A}) + (\boldsymbol{u}\times\operatorname{curl}\boldsymbol{A},\nabla\psi) - (\boldsymbol{v}\times\operatorname{curl}\boldsymbol{A},\nabla\varphi)\big\}\\ &+ \tfrac{\mu_0}{2}\big[(\boldsymbol{A}\times\operatorname{curl}\boldsymbol{u},\nabla\psi) - (\boldsymbol{A}\times\operatorname{curl}\boldsymbol{v},\nabla\varphi)\big] + \mu_0\big[(\boldsymbol{\omega}\times\boldsymbol{A},\nabla\psi) - (\boldsymbol{s}\times\boldsymbol{A},\nabla\varphi)\big]\\ &+ \tfrac{1}{2}\big[(\boldsymbol{\epsilon}\times\boldsymbol{F},\operatorname{curl}\boldsymbol{m}) - (\boldsymbol{\epsilon}\times\boldsymbol{m},\operatorname{curl}\boldsymbol{F})\big] + \tfrac{\mu_0}{2}\big[(\Delta\psi,\boldsymbol{u}\cdot\boldsymbol{A}) - (\Delta\varphi,\boldsymbol{v}\cdot\boldsymbol{A})\big].\end{aligned} \tag{3.6}$$

Lemma 3.1. *Under* (2.5)*, the forms are well defined on* $\mathbb{V}$*. The form* $\mathtt{A}$ *is bounded, symmetric, and coercive. The form* $\mathtt{J}$ *is bounded and skew-symmetric, and*

$$\mathtt{B}(\boldsymbol{z},\boldsymbol{x},\boldsymbol{y}) = -\mathtt{B}(\boldsymbol{z},\boldsymbol{y},\boldsymbol{x}), \qquad |\mathtt{B}(\boldsymbol{z},\boldsymbol{x},\boldsymbol{y})| \le C\|\boldsymbol{z}\|_{\mathbb{V}}\|\boldsymbol{x}\|_{\mathbb{V}}\|\boldsymbol{y}\|_{\mathbb{V}}. \tag{3.7}$$

The pressure form is bounded and satisfies

$$\sup_{\boldsymbol{y}\in\mathbb{V}\setminus\{0\}} \frac{|\mathtt{C}(\boldsymbol{y},q)|}{\|\boldsymbol{y}\|_{\mathbb{V}}} \ge c\|q\|, \qquad q\in W. \tag{3.8}$$

Proof. Completing a square gives

$$\begin{aligned}\mathtt{A}(\boldsymbol{y},\boldsymbol{y}) = {}& \eta\|\nabla\boldsymbol{v}\|^2 + \eta'\|\nabla\boldsymbol{s}\|^2 + (\eta'+\lambda')\|\operatorname{div}\boldsymbol{s}\|^2 + \zeta\|\operatorname{curl}\boldsymbol{v} - 2\boldsymbol{s}\|^2\\ &+ \sigma\big(\|\operatorname{curl}\boldsymbol{F}\|^2 + \|\operatorname{div}\boldsymbol{F}\|^2\big) + \tau^{-1}\|\boldsymbol{F}\|^2 + \alpha\|\nabla\psi\|^2 + \sigma\mu_0\|\Delta\psi\|^2.\end{aligned} \tag{3.9}$$

Every coefficient needed in (2.4) is positive. Thus $c\|\boldsymbol{y}\|_{\mathbb{V}}^2 \le \mathtt{A}(\boldsymbol{y},\boldsymbol{y}) \le C\|\boldsymbol{y}\|_{\mathbb{V}}^2$. Symmetry and boundedness follow directly. The two terms of $\mathtt{J}$ change sign when the arguments are exchanged. Each bracket in (3.6) has the same property. The remaining cross-product terms are also skew because $(\boldsymbol{a}\times\boldsymbol{b},\boldsymbol{c}) = -(\boldsymbol{a}\times\boldsymbol{c},\boldsymbol{b})$.

For continuity, (2.5) places the magnetic vector fields and the potential gradients in H^1. Use $H^1(\Omega)\hookrightarrow L^6(\Omega)$, interpolation into L^3 or L^4, and Hölder's inequality. For example,

$$|(\nabla\varphi,\boldsymbol{v}\operatorname{div}\boldsymbol{A})| \le \|\nabla\varphi\|_{L^3}\|\boldsymbol{v}\|_{L^6}\|\operatorname{div}\boldsymbol{A}\|, \qquad |(\Delta\psi,\boldsymbol{u}\cdot\boldsymbol{A})| \le \|\Delta\psi\|\|\boldsymbol{u}\|_{L^4}\|\boldsymbol{A}\|_{L^4}.$$

The other terms are estimated in the same way. Finally, for each $q\in L^2_0$ choose $\boldsymbol{v}\in H^1_0(\Omega)^3$ with $\operatorname{div}\boldsymbol{v} = q$ and $\|\nabla\boldsymbol{v}\| \le C\|q\|$. Testing the supremum with $(\boldsymbol{v},0,0,0)$ gives (3.8). □

Write $\|\boldsymbol{y}\|_{\mathtt{A}} = \mathtt{A}(\boldsymbol{y},\boldsymbol{y})^{1/2}$ and $\|\boldsymbol{y}\|_{\mathtt{D}} = \mathtt{D}(\boldsymbol{y},\boldsymbol{y})^{1/2}$. Let $\mathbb{H}_D$ be the completion of $\mathbb{V}$ in the $\mathtt{D}$ norm. For a linear functional ℓ, define

$$\|\ell\|_{\mathtt{A}^*} = \sup_{\boldsymbol{y}\neq0}\frac{|\langle\ell,\boldsymbol{y}\rangle|}{\|\boldsymbol{y}\|_{\mathtt{A}}}, \qquad \|\ell\|_{\mathtt{D}^*} = \sup_{\boldsymbol{y}\neq0}\frac{|\langle\ell,\boldsymbol{y}\rangle|}{\|\boldsymbol{y}\|_{\mathtt{D}}}. \tag{3.10}$$

The second quantity need not be finite for an arbitrary member of $\mathbb{V}'$.

3.2 Evolution problem and continuous energy balance

For smooth vector fields, the identity

$$\begin{aligned}(\boldsymbol{u}\cdot\nabla)\boldsymbol{m} = {}& \frac{1}{2}\boldsymbol{u}\operatorname{div}\boldsymbol{m} - \frac{1}{2}\boldsymbol{m}\operatorname{div}\boldsymbol{u} + \frac{1}{2}\nabla(\boldsymbol{u}\cdot\boldsymbol{m})\\ &+ \frac{1}{2}\operatorname{curl}\boldsymbol{u}\times\boldsymbol{m} + \frac{1}{2}\operatorname{curl}\boldsymbol{m}\times\boldsymbol{u} + \frac{1}{2}\operatorname{curl}(\boldsymbol{m}\times\boldsymbol{u})\end{aligned} \tag{3.11}$$

rewrites magnetization transport in a form suitable for curl–div spaces. The same identity, with $(\boldsymbol{u},\boldsymbol{m})$ replaced by $(\boldsymbol{m},\boldsymbol{H})$, rewrites the magnetic force. Introduce the shifted pressure

$$\widetilde{p} = p - \frac{\mu_0}{2}\boldsymbol{m}\cdot\nabla\varphi - \frac{1}{|\Omega|}\int_\Omega\Big(p - \frac{\mu_0}{2}\boldsymbol{m}\cdot\nabla\varphi\Big)\,\mathrm{d}x. \tag{3.12}$$

The weak formulation of (2.1), with the magnetostatic relation replaced by (2.9), reads as follows: find $\boldsymbol{x}(t) \in \mathbb{V}$ and $\widetilde{p}(t) \in W$ such that

$$\begin{aligned} \mathtt{D}(\partial_t \boldsymbol{x}, \boldsymbol{y}) + \mathtt{A}(\boldsymbol{x}, \boldsymbol{y}) + \mathtt{J}(\boldsymbol{x}, \boldsymbol{y}) + \mathtt{B}(\boldsymbol{x}, \boldsymbol{x}, \boldsymbol{y}) + \mathtt{C}(\boldsymbol{y}, \widetilde{p}) &= \langle \mathtt{f}, \boldsymbol{y} \rangle, \quad \boldsymbol{y} \in \mathbb{V}, \\ \mathtt{C}(\boldsymbol{x}, q) &= 0, \qquad q \in W, \end{aligned} \tag{3.13}$$

with $\boldsymbol{x}(0) = (\boldsymbol{u}_0, \boldsymbol{\omega}_0, \boldsymbol{m}_0, \varphi_0)$ and $\mathtt{f} = \operatorname{div} \partial_t \boldsymbol{H}_e + \beta \operatorname{div} \boldsymbol{H}_e - \sigma \Delta \operatorname{div} \boldsymbol{H}_e$.

Theorem 3.2 (Continuous energy estimate)**.** *Suppose that $\boldsymbol{x}$ is regular enough to be used as a test function and to satisfy the chain rule for the $\mathtt{D}$ norm, and that $\mathtt{f} \in L^1(0, T; \mathbb{H}'_D)$. Then, for $0 \le t \le T$,*

$$\|\boldsymbol{x}(t)\|_{\mathtt{D}}^2 + 2 \int_0^t \|\boldsymbol{x}(s)\|_{\mathtt{A}}^2 \, \mathrm{d}s = \|\boldsymbol{x}(0)\|_{\mathtt{D}}^2 + 2 \int_0^t \langle \mathtt{f}(s), \boldsymbol{x}(s) \rangle \, \mathrm{d}s. \tag{3.14}$$

In particular,

$$\sup_{0 \le t \le T} \left[\|\boldsymbol{x}(t)\|_{\mathtt{D}}^2 + 2 \int_0^t \|\boldsymbol{x}(s)\|_{\mathtt{A}}^2 \, \mathrm{d}s \right]^{1/2} \le \|\boldsymbol{x}(0)\|_{\mathtt{D}} + 2 \int_0^T \|\mathtt{f}(s)\|_{\mathtt{D}^*} \, \mathrm{d}s. \tag{3.15}$$

Proof. Set $\boldsymbol{y} = \boldsymbol{x}$ and $q = \widetilde{p}$ in (3.13). Both skew terms and the pressure term vanish. Therefore

$$\tfrac{1}{2} \tfrac{\mathrm{d}}{\mathrm{d}t} \|\boldsymbol{x}\|_{\mathtt{D}}^2 + \|\boldsymbol{x}\|_{\mathtt{A}}^2 = \langle \mathtt{f}, \boldsymbol{x} \rangle .$$

Integration gives (3.14). If M denotes the supremum on the left of (3.15) and $F = \int_0^T \|\mathtt{f}\|_{\mathtt{D}^*} \, \mathrm{d}s$, then $\sup_t \|\boldsymbol{x}(t)\|_{\mathtt{D}} \le M$ and $M^2 \le \|\boldsymbol{x}(0)\|_{\mathtt{D}}^2 + 2FM$. The elementary quadratic inequality gives $M \le \|\boldsymbol{x}(0)\|_{\mathtt{D}} + 2F$. □

Remark 3.3. The supremum in (3.15) is taken over the entire cumulative expression. It cannot be replaced, with the same right-hand side, by $\sup_t \|\boldsymbol{x}(t)\|_{\mathtt{D}}^2 + 2 \int_0^T \|\boldsymbol{x}\|_{\mathtt{A}}^2$. With zero forcing, the first term already contains the initial energy, so adding all later dissipation would generally exceed it. Also, zero forcing implies energy dissipation, not constancy of the state norm.

3.3 A nonlinear estimate used in the error analysis

To state sufficient regularity explicitly, set

$$\|\boldsymbol{x}\|_{\mathcal{R}} = \|\boldsymbol{u}\|_{W^{1,\infty}} + \|\boldsymbol{\omega}\|_{W^{1,\infty}} + \|\boldsymbol{m}\|_{W^{1,\infty}} + \|\varphi\|_{W^{2,\infty}}, \qquad \mathcal{R} = \{\boldsymbol{x} \in \mathbb{V} : \|\boldsymbol{x}\|_{\mathcal{R}} < \infty\}. \tag{3.16}$$

This is a sufficient smoothness class, not a claimed minimal one.

Lemma 3.4 (Estimate with one smooth argument)**.** *For $\boldsymbol{x} \in \mathcal{R}$ and $\boldsymbol{z}, \boldsymbol{y} \in \mathbb{V}$,*

$$|\mathtt{B}(\boldsymbol{z}, \boldsymbol{x}, \boldsymbol{y})| \le C \|\boldsymbol{x}\|_{\mathcal{R}} \|\boldsymbol{z}\|_{\mathtt{D}} \|\boldsymbol{y}\|_{\mathtt{A}}. \tag{3.17}$$

Proof. Derivatives on the first argument must be transferred before the estimate is applied. For example, the zero normal trace of $\boldsymbol{A}$ gives

$$(\nabla\varphi, \boldsymbol{v} \operatorname{div} \boldsymbol{A}) = -(\boldsymbol{A}, \nabla(\boldsymbol{v} \cdot \nabla\varphi)), \qquad (\nabla\psi, \boldsymbol{u} \operatorname{div} \boldsymbol{A}) = -(\boldsymbol{A}, \nabla(\boldsymbol{u} \cdot \nabla\psi)).$$

Both right-hand sides are bounded by $C\|\boldsymbol{x}\|_{\mathcal{R}} \|\boldsymbol{A}\| \|\boldsymbol{y}\|_{\mathtt{A}}$, using (2.5). Similarly,

$$(\boldsymbol{u} \times \operatorname{curl} \boldsymbol{A}, \nabla\psi) = (\boldsymbol{A}, \operatorname{curl}(\nabla\psi \times \boldsymbol{u})),$$

where the boundary term is zero because $\boldsymbol{u} = 0$. The analogous term containing $\boldsymbol{v}$ is treated in the same way. In $(\boldsymbol{m} \times \operatorname{curl} \boldsymbol{\epsilon}, \boldsymbol{F})$, transfer the curl to $\boldsymbol{F} \times \boldsymbol{m}$; $\boldsymbol{\epsilon} = 0$ removes the boundary term. Then (2.5) controls $\boldsymbol{F}$ in H^1.

The terms without derivatives on $\boldsymbol{z}$ use only its L^2 components. For instance,

$$|(\Delta\psi, \boldsymbol{u} \cdot \boldsymbol{A})| \le \|\boldsymbol{u}\|_{L^\infty} \|\boldsymbol{A}\| \|\Delta\psi\|, \quad |(\Delta\varphi, \boldsymbol{v} \cdot \boldsymbol{A})| \le \|\Delta\varphi\|_{L^\infty} \|\boldsymbol{A}\| \|\boldsymbol{v}\|.$$

The transport and remaining cross-product terms are bounded by the same argument. Summing these estimates proves (3.17). □

4 Arbitrary-order linearly implicit time discretization

This section discretizes time only. All states remain in the continuous space $\mathbb{V}$, and all pressures remain in W. We use the forms in Section 3; no mesh, finite element space, or discrete spatial operator is introduced here. The boundary closure and the compatibility conditions in Section 2.3 remain in force.

We combine extrapolation with algebraically stable Runge–Kutta methods. Related high-order constructions for gradient flows are studied in [1,8]. The present energy is already a quadratic form, so no additional energy variable is needed. We first prove solvability and an energy identity in the continuous spaces. We then give explicit stage conditions that imply arbitrary-order temporal convergence. The number of stages is chosen to meet these conditions, rather than inferred from the classical order of an ordinary differential equation method alone.

4.1 The time-semidiscrete scheme

Let

$$\mathfrak{K} = \{\boldsymbol{y} \in \mathbb{V} : \mathtt{C}(\boldsymbol{y}, q) = 0 \text{ for all } q \in W\}, \qquad E(\boldsymbol{x}) = \tfrac{1}{2}\mathtt{D}(\boldsymbol{x}, \boldsymbol{x}). \tag{4.1}$$

The kernel is a closed subspace of $\mathbb{V}$. Recall that $\mathtt{A}$ is symmetric and coercive, whereas $\mathtt{J}$ and the last two arguments of $\mathtt{B}$ are skew-symmetric. In particular, $\mathtt{J}(\boldsymbol{y}, \boldsymbol{y}) = \mathtt{B}(\boldsymbol{z}, \boldsymbol{y}, \boldsymbol{y}) = 0$ for all $\boldsymbol{y}, \boldsymbol{z} \in \mathbb{V}$.

Fix an extrapolation order $p \geq 1$. Let $\mathsf{A} = (a_{ij}) \in \mathbb{R}^{s\times s}$, $\boldsymbol{b} = (b_1, \dots, b_s)^{\mathsf{T}}$, and $\boldsymbol{c} = (c_1, \dots, c_s)^{\mathsf{T}}$ be the coefficients of an s-stage Runge–Kutta method. They are written in the Butcher tableau

$$\begin{array}{c|cccc} c_1 & a_{11} & a_{12} & \cdots & a_{1s} \\ c_2 & a_{21} & a_{22} & \cdots & a_{2s} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ c_s & a_{s1} & a_{s2} & \cdots & a_{ss} \\ \hline & b_1 & b_2 & \cdots & b_s \end{array}. \tag{4.2}$$

Here c_i specifies the stage time $t_n + c_i k$, a_{ij} determines the contribution of the jth stage derivative to the ith stage state, and b_i is the weight used in the endpoint update. The stage number s is chosen separately from the extrapolation order p.

For solvability and energy stability we require only

$$\mathsf{A} \text{ invertible}, \qquad b_i > 0, \qquad 0 \leq c_i \leq 1, \qquad \boldsymbol{c} = \mathsf{A}\mathbf{1}, \qquad \mathsf{G} \succeq 0, \tag{4.3}$$

where

$$\mathsf{G} = (g_{ij}), \qquad g_{ij} = b_i a_{ij} + b_j a_{ji} - b_i b_j. \tag{4.4}$$

The last condition, together with nonnegative weights, is algebraic stability [10]. Gauss and Radau IIA methods meet (4.3); their classical orders are $2s$ and $2s-1$, respectively. Gauss methods have $\mathsf{G} = 0$. Accuracy requirements beyond (4.3) will be stated separately.

Write $\Delta t = T/N$, $t_n = n\Delta t$, and assume $N \geq p$. Given $\boldsymbol{x}^0, \dots, \boldsymbol{x}^{p-1} \in \mathfrak{K}$, set, for $n = p-1, \dots, N-1$,

$$\widehat{\boldsymbol{x}}_{n,i} = \sum_{j=0}^{p-1} \ell_j(c_i)\boldsymbol{x}^{n-j}, \qquad \ell_j(c) = \prod_{\substack{0\leq m\leq p-1 \\ m\neq j}} \frac{c+m}{m-j}, \qquad t_{n,i} = t_n + c_i\Delta t. \tag{4.5}$$

For $p = 1$ the empty product equals one. Find stage states $\boldsymbol{x}_{n,i} \in \mathbb{V}$, stage derivatives $\mathcal{K}_{n,i} \in \mathbb{V}$,

and pressures $P_{n,i} \in W$ such that

$$\boldsymbol{x}_{n,i} = \boldsymbol{x}^n + \Delta t \sum_{j=1}^{s} a_{ij} \mathcal{K}_{n,j}, \tag{4.6a}$$

$$\mathtt{D}(\mathcal{K}_{n,i}, \boldsymbol{y}) + \mathtt{A}(\boldsymbol{x}_{n,i}, \boldsymbol{y}) + \mathtt{J}(\boldsymbol{x}_{n,i}, \boldsymbol{y}) + \mathtt{B}(\widehat{\boldsymbol{x}}_{n,i}, \boldsymbol{x}_{n,i}, \boldsymbol{y}) + \mathtt{C}(\boldsymbol{y}, P_{n,i}) = \langle \mathtt{f}(t_{n,i}), \boldsymbol{y} \rangle \,, \tag{4.6b}$$

$$\mathtt{C}(\boldsymbol{x}_{n,i}, q) = 0, \qquad i = 1, \dots, s, \tag{4.6c}$$

$$\boldsymbol{x}^{n+1} = \boldsymbol{x}^n + \Delta t \sum_{i=1}^{s} b_i \mathcal{K}_{n,i}. \tag{4.6d}$$

The equations hold for all $\boldsymbol{y} \in \mathbb{V}$ and $q \in W$. The first argument of $\mathtt{B}$ is known, while its second argument is an unknown stage state. Consequently, each step is a coupled linear variational problem. The pressures are independent stage multipliers, not averages of endpoint pressures.

4.2 Solvability and the energy identity

Theorem 4.1 (Unique solvability in the continuous spaces)**.** *Assume Lemma 3.1 and* (4.3)*. Given $\boldsymbol{x}^n \in \mathfrak{K}$, predictors $\widehat{\boldsymbol{x}}_{n,i} \in \mathbb{V}$, and $\mathtt{f}(t_{n,i}) \in \mathbb{V}'$, problem* (4.6) *has a unique solution for every $k > 0$. Its stage derivatives and updated state belong to $\mathfrak{K}$.*

Proof. Put $\mathsf{S} = \mathsf{A}^{-1}$ and $\mathsf{W}_b = \operatorname{diag}(b_1, \dots, b_s)$. Eliminating the stage derivatives gives, on $\mathfrak{K}^s = \mathfrak{K} \times \cdots \times \mathfrak{K}$, Equation (4.6) becomes: Find $\mathbf{U} = (\boldsymbol{x}_{n,1}, \dots, \boldsymbol{x}_{n,s})^{\mathsf{T}} \in \mathfrak{K}^s$, such that

$$a_n(\mathbf{U}, \mathbf{V}) = L_n(\mathbf{V}), \qquad \forall\ \mathbf{V} \in \mathfrak{K}^s,$$

with the bilinear form

$$a_n(\mathbf{U}, \mathbf{V}) = \frac{1}{\Delta t} \sum_{i,j=1}^{s} b_i S_{ij} \mathtt{D}(U_j, V_i) + \sum_{i=1}^{s} b_i \big[\mathtt{A}(U_i, V_i) + \mathtt{J}(U_i, V_i) + \mathtt{B}(\widehat{\boldsymbol{x}}_{n,i}, U_i, V_i)\big], \quad U_j = \boldsymbol{x}_{n,j} \tag{4.7}$$

and the right-hand side

$$L_n(\mathbf{V}) = \sum_{i=1}^{s} b_i \langle \mathtt{f}(t_{n,i}), V_i \rangle + \frac{1}{\Delta t} \sum_{i=1}^{s} b_i (\mathsf{S}\mathbf{1})_i \mathtt{D}(\boldsymbol{x}^n, V_i).$$

Both the form and the load are bounded on $\mathfrak{K}^s$ for the prescribed predictors. The identity

$$\frac{\mathsf{W}_b \mathsf{S} + \mathsf{S}^{\mathsf{T}} \mathsf{W}_b}{2} = \frac{1}{2} \mathsf{A}^{-\mathsf{T}} (\mathsf{G} + \boldsymbol{b}\boldsymbol{b}^{\mathsf{T}}) \mathsf{A}^{-1} \succeq 0 \tag{4.8}$$

shows that the mass term in $a_n(\mathbf{U}, \mathbf{U})$ is nonnegative. The skew terms vanish, so

$$a_n(\mathbf{U}, \mathbf{U}) \geq \sum_{i=1}^{s} b_i \|U_i\|_{\mathtt{A}}^2 \geq c \min_i b_i \sum_{i=1}^{s} \|U_i\|_{\mathbb{V}}^2.$$

Lax–Milgram therefore gives unique stages in $\mathfrak{K}^s$. Set

$$\mathcal{K}_{n,i} = \Delta t^{-1} \sum_{j=1}^{s} S_{ij} (\boldsymbol{x}_{n,j} - \boldsymbol{x}^n) \in \mathfrak{K}.$$

For each stage the remaining residual is a bounded functional on $\mathbb{V}$ that vanishes on $\mathfrak{K}$. The continuous inf-sup condition in (3.8) yields a unique pressure $P_{n,i} \in W$. Finally, (4.6d) preserves membership in $\mathfrak{K}$. □

Theorem 4.2 (Unconditional energy stability)**.** *Under the hypotheses of Theorem 4.1,*

$$\begin{aligned} &E(\boldsymbol{x}^{n+1}) - E(\boldsymbol{x}^n) + \Delta t \sum_{i=1}^{s} b_i \|\boldsymbol{x}_{n,i}\|_{\mathtt{A}}^2 \\ &\quad + \frac{\Delta t^2}{2} \sum_{i,j=1}^{s} g_{ij} \mathtt{D}(\mathcal{K}_{n,i}, \mathcal{K}_{n,j}) = \Delta t \sum_{i=1}^{s} b_i \left\langle \mathtt{f}(t_{n,i}), \boldsymbol{x}_{n,i} \right\rangle . \end{aligned} \tag{4.9}$$

In particular,

$$E(\boldsymbol{x}^{n+1}) + \frac{\Delta t}{2} \sum_{i=1}^{s} b_i \|\boldsymbol{x}_{n,i}\|_{\mathtt{A}}^2 \le E(\boldsymbol{x}^n) + \frac{\Delta t}{2} \sum_{i=1}^{s} b_i \|\mathtt{f}(t_{n,i})\|_{\mathtt{A}^*}^2. \tag{4.10}$$

If $\mathtt{f} = 0$*, then*

$$E(\boldsymbol{x}^{n+1}) + \Delta t \sum_{i=1}^{s} b_i \|\boldsymbol{x}_{n,i}\|_{\mathtt{A}}^2 \le E(\boldsymbol{x}^n). \tag{4.11}$$

For Gauss methods, i.e., $\mathsf{G} = 0$*,* (4.11) *is an equality.*

Proof. Test (4.6b) with $\boldsymbol{x}_{n,i}$. The skew terms and pressure term vanish, giving

$$\mathtt{D}(\mathcal{K}_{n,i}, \boldsymbol{x}_{n,i}) + \|\boldsymbol{x}_{n,i}\|_{\mathtt{A}}^2 = \left\langle \mathtt{f}(t_{n,i}), \boldsymbol{x}_{n,i} \right\rangle .$$

Expanding the quadratic energy of the update and using the stage relations yields

$$E(\boldsymbol{x}^{n+1}) - E(\boldsymbol{x}^n) = \Delta t \sum_{i=1}^{s} b_i \mathtt{D}(\mathcal{K}_{n,i}, \boldsymbol{x}_{n,i}) - \frac{\Delta t^2}{2} \sum_{i,j=1}^{s} g_{ij} \mathtt{D}(\mathcal{K}_{n,i}, \mathcal{K}_{n,j}). \tag{4.12}$$

This proves (4.9). The last sum is nonnegative: if $\mathsf{G} = \mathsf{R}^{\mathsf{T}}\mathsf{R}$, it equals $\sum_r \| \sum_i R_{ri} \mathcal{K}_{n,i} \|_{\mathtt{D}}^2$. Young's inequality gives (4.10), and zero forcing gives (4.11). □

Remark 4.3. Energy stability holds for every prescribed predictor in $\mathbb{V}$, regardless of its accuracy or the signs of its extrapolation coefficients. It does not imply arbitrary-order convergence by itself. Also, $\mathsf{G} = 0$ removes the additional Runge–Kutta dissipation, not the physical dissipation in $\mathtt{A}$.

4.3 Arbitrary-order temporal error estimates

We use sufficient stage and quadrature conditions that can be checked directly:

$$\sum_{j=1}^{s} a_{ij} c_j^{\ell-1} = \frac{c_i^{\ell}}{\ell} \quad (1 \le \ell \le p-1), \qquad \sum_{i=1}^{s} b_i c_i^{\ell-1} = \frac{1}{\ell} \quad (1 \le \ell \le p). \tag{4.13}$$

The first group is empty if $p = 1$. In the standard notation these conditions are $C(p-1)$ and $B(p)$ [1, 10]. They distinguish the temporal result below from an ordinary differential equation order statement. For an s-stage Gauss method, the present proof allows $p \le \min\{s+1, 2s\}$; for Radau IIA it allows $p \le \min\{s+1, 2s-1\}$. Thus backward Euler covers $p = 1$, and the $(p-1)$-stage Gauss method covers every $p \ge 2$. This gives a construction of arbitrary temporal order without a spatial discretization.

Theorem 4.4 (Order-p temporal convergence)**.** *Assume* (4.3)*,* (4.13)*, and Lemmas 3.1 and 3.4. Let* $\boldsymbol{x}$ *solve* (3.13)*, with*

$$\boldsymbol{x} \in C^{p+1}([0,T]; \mathbb{V}) \cap C^1([0,T]; \mathcal{R}), \tag{4.14}$$

where $\mathcal{R}$ *is defined in* (3.16)*. Assume that the equations hold at the stage times and that* $\boldsymbol{x}^0, \dots, \boldsymbol{x}^{p-1} \in \mathfrak{K}$*. Set* $\epsilon_{\mathrm{in}} = \max_{0 \le j < p} \|\boldsymbol{x}(t_j) - \boldsymbol{x}^j\|_{\mathtt{D}}$*. Then the solution of* (4.6) *satisfies*

$$\max_{0 \le n \le N} \|\boldsymbol{x}(t_n) - \boldsymbol{x}^n\|_{\mathtt{D}}^2 + \Delta t \sum_{n=p-1}^{N-1} \sum_{i} b_i \|\boldsymbol{x}(t_{n,i}) - \boldsymbol{x}_{n,i}\|_{\mathtt{A}}^2 \le C(\epsilon_{\mathrm{in}}^2 + \Delta t^{2p}). \tag{4.15}$$

The constant depends on the fixed coefficients, p*,* T*, the physical parameters, the domain, and the bounds in* (4.14)*, but not on* Δt*. In particular,* $\epsilon_{\mathrm{in}} = O(\Delta t^p)$ *gives order* p *in the stated norms.*

Proof. Write $x_n = \boldsymbol{x}(t_n)$ and $x_{n,i} = \boldsymbol{x}(t_{n,i})$. Introduce reference stages that satisfy the Runge–Kutta stage relations exactly:

$$U_{n,i} = \boldsymbol{x}(t_n) + \Delta t \sum_{j=1}^{s} a_{ij} \partial_t \boldsymbol{x}(t_{n,j}), \qquad d_n = \boldsymbol{x}(t_{n+1}) - \boldsymbol{x}(t_n) - \Delta t \sum_{i=1}^{s} b_i \partial_t \boldsymbol{x}(t_{n,i}). \tag{4.16}$$

They lie in $\mathfrak{K}$. Let $\widehat{\boldsymbol{x}}(t_{n,i}) = \sum_{j=0}^{p-1} \ell_j(c_i) \boldsymbol{x}(t_{n-j})$. Taylor expansion in $\mathbb{V}$, using (4.13), gives

$$\max_i \|U_{n,i} - \boldsymbol{x}(t_{n,i})\|_{\mathbb{V}} + \max_i \|\widehat{\boldsymbol{x}}(t_{n,i}) - \boldsymbol{x}(t_{n,i})\|_{\mathbb{V}} \le C\Delta t^p, \qquad \|d_n\|_{\mathtt{D}} \le C\Delta t^{p+1}. \tag{4.17}$$

Moreover (4.14) implies $\|U_{n,i}\|_{\mathcal{R}} \le C$.

Set $e^n = \boldsymbol{x}(t_n) - \boldsymbol{x}^n$, $E_{n,i} = U_{n,i} - \boldsymbol{x}_{n,i}$, $D_{n,i} = \partial_t \boldsymbol{x}(t_{n,i}) - \mathcal{K}_{n,i}$, and $\widehat{e}_{n,i} = \widehat{\boldsymbol{x}}(t_{n,i}) - \widehat{\boldsymbol{x}}_{n,i}$. Then

$$E_{n,i} = e^n + \Delta t \sum_{j=1}^{s} a_{ij} D_{n,j} \in \mathfrak{K}, \qquad e^{n+1} - d_n = e^n + \Delta t \sum_{i=1}^{s} b_i D_{n,i} \in \mathfrak{K}.$$

Subtraction of the numerical and reference equations with test function $\boldsymbol{y} \in \mathfrak{K}$, gives

$$\mathtt{D}(D_{n,i}, \boldsymbol{y}) + \mathtt{A}(E_{n,i}, \boldsymbol{y}) + \mathtt{J}(E_{n,i}, \boldsymbol{y}) + \mathtt{B}(\widehat{\boldsymbol{x}}_{n,i}, E_{n,i}, \boldsymbol{y}) = R_{n,i}(\boldsymbol{y}) - \mathtt{B}(\widehat{e}_{n,i}, U_{n,i}, \boldsymbol{y}), \tag{4.18}$$

whith

$$\begin{aligned} R_{n,i}(\boldsymbol{y}) = {} & \mathtt{A}(U_{n,i} - \boldsymbol{x}(t_{n,i}), \boldsymbol{y}) + \mathtt{J}(U_{n,i} - \boldsymbol{x}(t_{n,i}), \boldsymbol{y}) \\ & + \mathtt{B}(\widehat{\boldsymbol{x}}(t_{n,i}), U_{n,i}, \boldsymbol{y}) - \mathtt{B}(\boldsymbol{x}(t_{n,i}), \boldsymbol{x}(t_{n,i}), \boldsymbol{y}). \end{aligned} \tag{4.19}$$

Split the last line as

$$\mathtt{B}(\widehat{\boldsymbol{x}}(t_{n,i}), U_{n,i} - \boldsymbol{x}(t_{n,i}), \boldsymbol{y}) + \mathtt{B}(\widehat{\boldsymbol{x}}(t_{n,i}) - \boldsymbol{x}(t_{n,i}), \boldsymbol{x}(t_{n,i}), \boldsymbol{y}).$$

Equation (4.17) implies

$$\|R_{n,i}\|_{\mathtt{A}^*} \le C\Delta t^p. \tag{4.20}$$

Take $\boldsymbol{y} = E_{n,i} \in \mathfrak{K}$ in (4.18) and apply the quadratic identity (4.12) to these error relations. Drop the nonnegative G term and use Lemma 3.4 with the smooth argument $U_{n,i}$. If $M_n = \max_{0 \le j \le n} \|e^j\|_{\mathtt{D}}$, then $\|\widehat{e}_{n,i}\|_{\mathtt{D}} \le CM_n$, and Young's inequality gives

$$\|e^{n+1} - d_n\|_{\mathtt{D}}^2 + \Delta t \sum_{i=1}^{s} b_i \|E_{n,i}\|_{\mathtt{A}}^2 \le \|e^n\|_{\mathtt{D}}^2 + C\Delta t M_n^2 + C\Delta t^{2p+1}. \tag{4.21}$$

Using the following inequality

$$\|e^{n+1}\|_{\mathtt{D}}^2 \le (1 + \Delta t)\|e^{n+1} - d_n\|_{\mathtt{D}}^2 + (1 + \Delta t^{-1})\|d_n\|_{\mathtt{D}}^2.$$

Since $\Delta t \le T$, (4.17) and (4.21) yield

$$\|e^{n+1}\|_{\mathtt{D}}^2 - \|e^n\|_{\mathtt{D}}^2 + \Delta t \sum_{i=1}^{s} b_i \|E_{n,i}\|_{\mathtt{A}}^2 \le C\Delta t M_n^2 + C\Delta t^{2p+1}. \tag{4.22}$$

Thus $M_{n+1}^2 \le (1 + C\Delta t) M_n^2 + C\Delta t^{2p+1}$, and iteration bounds the endpoint error by $C(\epsilon_{in}^2 + \Delta t^{2p})$. Summing (4.22) bounds the accumulated stage error. Finally, $\boldsymbol{x}(t_{n,i}) - \boldsymbol{x}_{n,i} = E_{n,i} + \boldsymbol{x}(t_{n,i}) - U_{n,i}$ and (4.17) give (4.15). □

4.4 Starting values

One sufficient linear start uses time derivatives of compatible initial data. Suppose $\partial_t^k \boldsymbol{x}(0) \in \mathfrak{K}$ are available for $0 \le k \le p-1$. In each of the first $p-1$ steps, replace (4.5) by the prescribed polynomial

$$Z_{n,i}^{\mathrm{start}} = \sum_{k=0}^{p-1} \frac{t_{n,i}^k}{k!} \partial_t^k \boldsymbol{x}(0), \qquad 0 \le n \le p-2, \tag{4.23}$$

and otherwise use (4.6). For $p = 1$ no starting step is needed.

Proposition 4.5 (A time-only linear start)**.** *Under Theorem 4.4, let* $\boldsymbol{x}^0 = \boldsymbol{x}(0)$ *and use* (4.23) *in the first* $p-1$ *steps. Then the resulting starting values satisfy* $\epsilon_{\mathrm{in}} \le C\Delta t^p$*. Every starting step is linear, uniquely solvable, and satisfies* (4.9)*.*

Proof. Taylor's formula gives $\|Z_{n,i}^{\mathrm{start}} - \boldsymbol{x}(t_{n,i})\|_{\mathbb{V}} \le C\Delta t^p$, since $0 \le t_{n,i} \le (p-1)\Delta t$ and p is fixed. In the proof of Theorem 4.4, use this same prescribed predictor for the reference and numerical starting equations. The predictor-error term in (4.18) is then absent, and the residual remains $O(\Delta t^p)$ in $\mathtt{A}^*$ norm. The same energy recurrence, starting with $e^0 = 0$, gives the stated bound over the first $p-1$ steps. Solvability and the energy identity follow from Theorems 4.1 and 4.2. □

The derivatives in this construction are a compatibility requirement on the initial data, not extra unknown exact values at later times. When they exist, they can be characterized recursively, for $0 \le k \le p-2$, by differentiating (3.13):

$$\begin{aligned} \mathtt{D}(\partial_t^{k+1}\boldsymbol{x}(0), \boldsymbol{y}) + \mathtt{C}(\boldsymbol{y}, \partial_t^k \widetilde{p}(0)) = {} & \left\langle \partial_t^k \mathtt{f}(0), \boldsymbol{y} \right\rangle - \mathtt{A}(\partial_t^k \boldsymbol{x}(0), \boldsymbol{y}) - \mathtt{J}(\partial_t \boldsymbol{x}(0), \boldsymbol{y}) \\ & - \sum_{r=0}^{k} \binom{k}{r} \mathtt{B}(\partial_t^r \boldsymbol{x}(0), \partial_t^{k-r} \boldsymbol{x}(0), \boldsymbol{y}), \qquad \forall\, \boldsymbol{y} \in \mathbb{V}, \\ \mathtt{C}(\partial_t^{k+1}\boldsymbol{x}(0), q) = {} & 0, \qquad \forall\, q \in W. \end{aligned} \tag{4.24}$$

The right-hand side uses only lower derivatives. Membership of these derivatives in $\mathbb{V}$ is not automatic for general data. The convergence theorem can instead be used with any supplied order-p starting values.

4.5 A first-order example: the semi-implicit backward Euler scheme

The first-order semi-implicit scheme is contained in the general Runge–Kutta framework (4.6). It is obtained from the one-stage Radau IIA method by choosing

$$s = 1, \qquad p = 1, \qquad \mathsf{A} = (a_{11}) = (1), \qquad \boldsymbol{b} = (b_1) = (1), \qquad \boldsymbol{c} = (c_1) = (1). \tag{4.25}$$

The corresponding Butcher tableau is

$$\begin{array}{c|c} 1 & 1 \\ \hline & 1 \end{array}. \tag{4.26}$$

Here $s = 1$ means that there is one stage, while $p = 1$ is the extrapolation order and the resulting temporal order.

For $p = 1$, the predictor (4.5) contains only the term $j = 0$. Since the corresponding product is empty,

$$\ell_0(c) = 1.$$

Therefore,

$$\widehat{\boldsymbol{x}}_{n,1} = \boldsymbol{x}^n. \tag{4.27}$$

Moreover,

$$t_{n,1} = t_n + c_1 \Delta t = t_{n+1}.$$

The matrix in the algebraic stability condition is

$$\mathsf{G} = (g_{11}), \qquad g_{11} = 2b_1a_{11} - b_1^2 = 1, \tag{4.28}$$

and hence

$$\mathsf{G} = (1) \succeq 0.$$

Thus all assumptions in (4.3) are satisfied. The stage condition $C(p-1) = C(0)$ is empty, while the quadrature condition $B(1)$ reduces to $b_1 = 1$.

With the coefficients in (4.25), the stage relation and the endpoint update in (4.6) become

$$\boldsymbol{x}_{n,1} = \boldsymbol{x}^n + \Delta t \mathcal{K}_{n,1}, \qquad \boldsymbol{x}^{n+1} = \boldsymbol{x}^n + \Delta t \mathcal{K}_{n,1}.$$

Consequently,

$$\boldsymbol{x}_{n,1} = \boldsymbol{x}^{n+1}, \qquad \mathcal{K}_{n,1} = \frac{\boldsymbol{x}^{n+1} - \boldsymbol{x}^n}{\Delta t} = \delta_t \boldsymbol{x}^{n+1}, \qquad P_{n,1} = \widetilde{p}^{n+1}. \tag{4.29}$$

For $0 \le n \le N-1$, substitution of (4.27)–(4.29) into the general stage equation gives

$$\begin{aligned} \mathtt{D}(\delta_t \boldsymbol{x}^{n+1}, \boldsymbol{y}) + \mathtt{A}(\boldsymbol{x}^{n+1}, \boldsymbol{y}) + \mathtt{J}(\boldsymbol{x}^{n+1}, \boldsymbol{y}) + \mathtt{B}(\boldsymbol{x}^n, \boldsymbol{x}^{n+1}, \boldsymbol{y}) + \mathtt{C}(\boldsymbol{y}, \widetilde{p}^{n+1}) &= \langle \mathtt{f}(t_{n+1}), \boldsymbol{y} \rangle, \quad \forall\, \boldsymbol{y} \in \mathbb{V}, \\ \mathtt{C}(\boldsymbol{x}^{n+1}, q) &= 0, \qquad \forall\, q \in W, \end{aligned} \tag{4.30}$$

where

$$\delta_t \boldsymbol{x}^n = \frac{\boldsymbol{x}^n - \boldsymbol{x}^{n-1}}{\Delta t}.$$

4.6 A second-order example: the extrapolated Crank–Nicolson scheme

The second-order extrapolated Crank–Nicolson scheme is a special case of the general Runge–Kutta formulation (4.6). It is obtained from the one-stage Gauss method by choosing

$$s = 1, \qquad p = 2, \qquad \mathsf{A} = (a_{11}) = \left(\frac{1}{2}\right), \qquad \boldsymbol{b} = (b_1) = (1), \qquad \boldsymbol{c} = (c_1) = \left(\frac{1}{2}\right). \tag{4.31}$$

The corresponding Butcher tableau is

$$\begin{array}{c|c} \frac{1}{2} & \frac{1}{2} \\ \hline & 1 \end{array}. \tag{4.32}$$

The underlying Runge–Kutta method is the implicit midpoint rule. It has classical order two and stage order one.

For $p = 2$, the extrapolation basis functions in (4.5) are

$$\ell_0(c) = c + 1, \qquad \ell_1(c) = -c.$$

Since $c_1 = 1/2$, the predictor is

$$\widehat{\boldsymbol{x}}_{n,1} = \ell_0\left(\frac{1}{2}\right)\boldsymbol{x}^n + \ell_1\left(\frac{1}{2}\right)\boldsymbol{x}^{n-1} = \frac{3}{2}\boldsymbol{x}^n - \frac{1}{2}\boldsymbol{x}^{n-1}. \tag{4.33}$$

The algebraic stability matrix contains only one entry:

$$\mathsf{G} = (g_{11}), \qquad g_{11} = 2b_1a_{11} - b_1^2 = 2 \cdot 1 \cdot \frac{1}{2} - 1 = 0. \tag{4.34}$$

Hence

$$\mathsf{G} = (0) \succeq 0.$$

The order conditions required for $p = 2$ also hold:

$$a_{11} = c_1 = \frac{1}{2}, \qquad b_1 = 1, \qquad b_1c_1 = \frac{1}{2}.$$

These are precisely the conditions $C(1)$ and $B(2)$.

With the coefficients (4.31), the stage relation and endpoint update in (4.6) become

$$\boldsymbol{x}_{n,1} = \boldsymbol{x}^n + \frac{\Delta t}{2}\mathcal{K}_{n,1}, \qquad \boldsymbol{x}^{n+1} = \boldsymbol{x}^n + \Delta t \mathcal{K}_{n,1}.$$

Therefore,

$$\boldsymbol{x}_{n,1} = \frac{\boldsymbol{x}^{n+1} + \boldsymbol{x}^n}{2}, \qquad \mathcal{K}_{n,1} = \frac{\boldsymbol{x}^{n+1} - \boldsymbol{x}^n}{\Delta t}, \qquad P_{n,1} = \widetilde{p}^{\,n+1/2}. \tag{4.35}$$

Introduce

$$\bar{\boldsymbol{x}}^{\,n+1} := \frac{\boldsymbol{x}^{n+1} + \boldsymbol{x}^n}{2}, \qquad \widehat{\boldsymbol{x}}^{\,n} := \frac{3}{2}\boldsymbol{x}^n - \frac{1}{2}\boldsymbol{x}^{n-1}, \qquad t_{n+1/2} := t_n + \frac{\Delta t}{2}. \tag{4.36}$$

Then

$$\boldsymbol{x}_{n,1} = \bar{\boldsymbol{x}}^{\,n+1}, \qquad \mathcal{K}_{n,1} = \delta_t \boldsymbol{x}^{n+1}, \qquad \widehat{\boldsymbol{x}}_{n,1} = \widehat{\boldsymbol{x}}^{\,n}.$$

For $0 \le n \le N-1$, substitution into the general stage equation (4.6) gives

$$\begin{aligned} &\mathtt{D}(\delta_t \boldsymbol{x}^{n+1}, \boldsymbol{y}) + \mathtt{A}(\bar{\boldsymbol{x}}^{\,n+1}, \boldsymbol{y}) + \mathtt{J}(\bar{\boldsymbol{x}}^{\,n+1}, \boldsymbol{y}) \\ &\qquad + \mathtt{B}(\widehat{\boldsymbol{x}}^{\,n}, \bar{\boldsymbol{x}}^{\,n+1}, \boldsymbol{y}) + \mathtt{C}(\boldsymbol{y}, \widetilde{p}^{\,n+1/2}) = \left\langle \mathtt{f}(t_{n+1/2}), \boldsymbol{y} \right\rangle, \quad \forall\ \boldsymbol{y} \in \mathbb{V}, \\ &\qquad\qquad \mathtt{C}(\bar{\boldsymbol{x}}^{\,n+1}, q) = 0, \qquad\qquad \forall\ q \in W. \end{aligned} \tag{4.37}$$

The first argument of $\mathtt{B}$ is known, while its second argument is the current midpoint state. Hence (4.37) is linear at every step once the starting value $\boldsymbol{x}^1$ is available. The pressure $\widetilde{p}^{\,n+1/2}$ is an independent midpoint multiplier; it is not defined as an average of endpoint pressures.

The two-step extrapolation requires a second-order accurate value $\boldsymbol{x}^1$. One possible choice is the nonlinear midpoint start: find $(\boldsymbol{x}^1, \widetilde{p}^{\,1/2}) \in \mathbb{V} \times W$ such that

$$\begin{aligned} \mathtt{D}(\delta_t \boldsymbol{x}^1, \boldsymbol{y}) + \mathtt{A}(\bar{\boldsymbol{x}}^{\,1}, \boldsymbol{y}) + \mathtt{J}(\bar{\boldsymbol{x}}^{\,1}, \boldsymbol{y}) + \mathtt{B}(\bar{\boldsymbol{x}}^{\,1}, \bar{\boldsymbol{x}}^{\,1}, \boldsymbol{y}) + \mathtt{C}(\boldsymbol{y}, \widetilde{p}^{\,1/2}) &= \left\langle \mathtt{f}(t_{1/2}), \boldsymbol{y} \right\rangle, \quad \boldsymbol{y} \in \mathbb{V}, \\ \mathtt{C}(\bar{\boldsymbol{x}}^{\,1}, q) &= 0, \qquad\qquad q \in W, \end{aligned} \tag{4.38}$$

where

$$\bar{\boldsymbol{x}}^{\,1} = \frac{\boldsymbol{x}^1 + \boldsymbol{x}^0}{2}.$$

The starting problem (4.38) is nonlinear. In the time-semidiscrete analysis, a solution is assumed to be available. Its existence after spatial discretization is treated separately. Alternatively, the linear starting procedure in Proposition 4.5 gives a second-order starting value when the required compatible initial derivative is available.

4.7 A third-order time-semidiscrete Gauss-type example

We now construct a third-order linearly implicit method within the general framework (4.6). The temporal convergence theorem with $p = 3$ requires the stage condition $C(2)$, the quadrature condition $B(3)$, and a third-order extrapolation. The minimal Gauss method satisfying these conditions is the two-stage Gauss method. We therefore choose

$$s = 2, \qquad p = 3. \tag{4.39}$$

The Runge–Kutta coefficients are

$$\mathsf{A} = \begin{pmatrix} \frac{1}{4} & \frac{1}{4} - \frac{\sqrt{3}}{6} \\ \frac{1}{4} + \frac{\sqrt{3}}{6} & \frac{1}{4} \end{pmatrix}, \qquad \boldsymbol{b} = \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \end{pmatrix}, \qquad \boldsymbol{c} = \begin{pmatrix} \frac{1}{2} - \frac{\sqrt{3}}{6} \\ \frac{1}{2} + \frac{\sqrt{3}}{6} \end{pmatrix}, \qquad \mathsf{G} = 0. \tag{4.40}$$

The two-stage Gauss method has classical order four and stage order two. Its coefficients satisfy

$$\mathsf{A}\mathbf{1} = \boldsymbol{c}, \qquad \mathsf{A}\boldsymbol{c} = \frac{1}{2}\begin{pmatrix} c_1^2 \\ c_2^2 \end{pmatrix}, \tag{4.41}$$

and

$$\boldsymbol{b}^{\mathsf{T}}\mathbf{1} = 1, \qquad \boldsymbol{b}^{\mathsf{T}}\boldsymbol{c} = \frac{1}{2}, \qquad \boldsymbol{b}^{\mathsf{T}}\begin{pmatrix} c_1^2 \\ c_2^2 \end{pmatrix} = \frac{1}{3}. \tag{4.42}$$

Thus the conditions $C(2)$ and $B(3)$ in (4.13) hold.

For $p = 3$, the Lagrange extrapolation basis functions are

$$\ell_0(c) = \frac{(c+1)(c+2)}{2}, \qquad \ell_1(c) = -c(c+2), \qquad \ell_2(c) = \frac{c(c+1)}{2}. \tag{4.43}$$

The stage predictors are therefore

$$\begin{aligned} \widehat{\boldsymbol{x}}_{n,1} &= \left(\frac{23}{12} - \frac{\sqrt{3}}{3}\right)\boldsymbol{x}^n + \left(-\frac{4}{3} + \frac{\sqrt{3}}{2}\right)\boldsymbol{x}^{n-1} + \left(\frac{5}{12} - \frac{\sqrt{3}}{6}\right)\boldsymbol{x}^{n-2}, \\ \widehat{\boldsymbol{x}}_{n,2} &= \left(\frac{23}{12} + \frac{\sqrt{3}}{3}\right)\boldsymbol{x}^n + \left(-\frac{4}{3} - \frac{\sqrt{3}}{2}\right)\boldsymbol{x}^{n-1} + \left(\frac{5}{12} + \frac{\sqrt{3}}{6}\right)\boldsymbol{x}^{n-2}. \end{aligned} \tag{4.44}$$

Given

$$\boldsymbol{x}^n, \boldsymbol{x}^{n-1}, \boldsymbol{x}^{n-2} \in \mathfrak{K},$$

find

$$\boldsymbol{x}_{n,i}, \mathcal{K}_{n,i} \in \mathbb{V}, \qquad P_{n,i} \in W, \qquad i = 1, 2,$$

such that

$$\boldsymbol{x}_{n,1} = \boldsymbol{x}^n + \Delta t\left[\frac{1}{4}\mathcal{K}_{n,1} + \left(\frac{1}{4} - \frac{\sqrt{3}}{6}\right)\mathcal{K}_{n,2}\right], \tag{4.45a}$$

$$\boldsymbol{x}_{n,2} = \boldsymbol{x}^n + \Delta t\left[\left(\frac{1}{4} + \frac{\sqrt{3}}{6}\right)\mathcal{K}_{n,1} + \frac{1}{4}\mathcal{K}_{n,2}\right], \tag{4.45b}$$

$$\mathtt{D}(\mathcal{K}_{n,i}, \boldsymbol{y}) + \mathtt{A}(\boldsymbol{x}_{n,i}, \boldsymbol{y}) + \mathtt{J}(\boldsymbol{x}_{n,i}, \boldsymbol{y}) + \mathtt{B}(\widehat{\boldsymbol{x}}_{n,i}, \boldsymbol{x}_{n,i}, \boldsymbol{y}) + \mathtt{C}(\boldsymbol{y}, P_{n,i}) = \langle \mathtt{f}(t_{n,i}), \boldsymbol{y}\rangle, \qquad \boldsymbol{y} \in \mathbb{V}, \tag{4.45c}$$

$$\mathtt{C}(\boldsymbol{x}_{n,i}, q) = 0, \qquad q \in W, \tag{4.45d}$$

$$\boldsymbol{x}^{n+1} = \boldsymbol{x}^n + \frac{\Delta t}{2}\left(\mathcal{K}_{n,1} + \mathcal{K}_{n,2}\right). \tag{4.45e}$$

4.8 A fourth-order time-semidiscrete example

We now specialize the general Runge–Kutta formulation (4.6) to a fourth-order linearly implicit method. To satisfy the stage condition $C(3)$ required by Theorem 4.4, we choose the three-stage Gauss method and set

$$s = 3, \qquad p = 4. \tag{4.46}$$

The corresponding coefficients are

$$\mathsf{A} = \begin{pmatrix} \dfrac{5}{36} & \dfrac{2}{9} - \dfrac{\sqrt{15}}{15} & \dfrac{5}{36} - \dfrac{\sqrt{15}}{30} \\ \dfrac{5}{36} + \dfrac{\sqrt{15}}{24} & \dfrac{2}{9} & \dfrac{5}{36} - \dfrac{\sqrt{15}}{24} \\ \dfrac{5}{36} + \dfrac{\sqrt{15}}{30} & \dfrac{2}{9} + \dfrac{\sqrt{15}}{15} & \dfrac{5}{36} \end{pmatrix}, \quad \boldsymbol{b} = \begin{pmatrix} 5/18 \\ 4/9 \\ 5/18 \end{pmatrix}, \quad \boldsymbol{c} = \begin{pmatrix} 1/2 - \sqrt{15}/10 \\ 1/2 \\ 1/2 + \sqrt{15}/10 \end{pmatrix}, \quad \mathsf{G} = 0. \tag{4.47}$$

The three-stage Gauss method has classical order six and stage order three. In particular, it satisfies $C(3)$ and $B(6)$, and hence also satisfies the conditions $C(3)$ and $B(4)$ required for the fourth-order result in Theorem 4.4 [10].

For $p = 4$, the extrapolated value at the ith stage is

$$\begin{aligned}\widehat{\boldsymbol{x}}_{n,i} = &\frac{(c_i+1)(c_i+2)(c_i+3)}{6}\,\boldsymbol{x}^n - \frac{c_i(c_i+2)(c_i+3)}{2}\,\boldsymbol{x}^{n-1} \\ &+ \frac{c_i(c_i+1)(c_i+3)}{2}\,\boldsymbol{x}^{n-2} - \frac{c_i(c_i+1)(c_i+2)}{6}\,\boldsymbol{x}^{n-3}, \qquad i = 1,2,3.\end{aligned} \tag{4.48}$$

Given

$$\boldsymbol{x}^n, \boldsymbol{x}^{n-1}, \boldsymbol{x}^{n-2}, \boldsymbol{x}^{n-3} \in \mathfrak{K},$$

find the stage states

$$\boldsymbol{x}_{n,i} \in \mathbb{V}, \qquad \mathcal{K}_{n,i} \in \mathbb{V}, \qquad P_{n,i} \in W, \qquad i = 1,2,3,$$

such that

$$\boldsymbol{x}_{n,i} = \boldsymbol{x}^n + \Delta t \sum_{j=1}^{3} a_{ij}\mathcal{K}_{n,j}, \qquad i = 1,2,3, \tag{4.49a}$$

$$\mathtt{D}(\mathcal{K}_{n,i}, \boldsymbol{y}) + \mathtt{A}(\boldsymbol{x}_{n,i}, \boldsymbol{y}) + \mathtt{J}(\boldsymbol{x}_{n,i}, \boldsymbol{y}) + \mathtt{B}(\widehat{\boldsymbol{x}}_{n,i}, \boldsymbol{x}_{n,i}, \boldsymbol{y}) + \mathtt{C}(\boldsymbol{y}, P_{n,i}) = \langle \mathtt{f}(t_{n,i}), \boldsymbol{y}\rangle\,, \qquad \boldsymbol{y} \in \mathbb{V}, \tag{4.49b}$$

$$\mathtt{C}(\boldsymbol{x}_{n,i}, q) = 0, \qquad q \in W, \qquad i = 1,2,3, \tag{4.49c}$$

$$\boldsymbol{x}^{n+1} = \boldsymbol{x}^n + \Delta t\left(\frac{5}{18}\mathcal{K}_{n,1} + \frac{4}{9}\mathcal{K}_{n,2} + \frac{5}{18}\mathcal{K}_{n,3}\right). \tag{4.49d}$$

Since the predictors $\widehat{\boldsymbol{x}}_{n,i}$ are known, the three stages form one coupled linear variational problem. They are not three independent one-stage problems.

5 Finite element discretization

5.1 Finite element spaces

Let $\mathcal{T}_h$ be a conforming shape-regular tetrahedral mesh. For any integer $l \geq 1$, use the compatible sequence

$$\Sigma_h \xrightarrow{\text{grad}} \boldsymbol{U}_h \xrightarrow{\text{curl}} \boldsymbol{V}_h \xrightarrow{\text{div}} W_h \tag{5.1}$$

with homogeneous traces as in the corresponding continuous spaces. Here Σ_h contains continuous degree-l scalar polynomials, $\boldsymbol{U}_h$ is the first-family Nédélec space of order $(l-1)$, $\boldsymbol{V}_h$ is the Raviart–Thomas space of order $(l-1)$, and W_h contains discontinuous degree-$(l-1)$ polynomials of zero mean. These spaces fit the finite element de Rham complex [3, 4, 16, 20].

For the velocity–pressure discretization, we employ stable mixed finite element pairs. For the lowest-order case $l = 1$, we use the mini element pair $\boldsymbol{S}_h \times Q_h$, where the velocity space consists of continuous piecewise linear functions enriched with element bubble functions, and the pressure space consists of continuous piecewise linear functions with zero mean [2]. For higher-order cases $l \geq 2$, we adopt the classical Taylor–Hood finite element pair [32]:

$$\boldsymbol{S}_h = \left\{\boldsymbol{v}_h \in [C^0(\overline{\Omega})]^d : \boldsymbol{v}_h|_K \in [\mathbb{P}_l(K)]^d,\ \boldsymbol{v}_h = 0 \text{ on } \partial\Omega\right\},$$

and

$$Q_h = \left\{q_h \in C^0(\overline{\Omega}) : q_h|_K \in \mathbb{P}_{l-1}(K),\ \int_\Omega q_h\,dx = 0\right\}.$$

The pair $(\boldsymbol{S}_h, Q_h)$ satisfies the discrete Ladyzhenskaya–Babuška–Brezzi (LBB) condition: there exists a constant $c_0 > 0$, independent of the mesh size h, such that

$$\inf_{q_h \in Q_h\setminus\{0\}} \sup_{\boldsymbol{v}_h \in \boldsymbol{S}_h\setminus\{0\}} \frac{|(\operatorname{div}\boldsymbol{v}_h, q_h)|}{\|\nabla \boldsymbol{v}_h\|_0 \|q_h\|_0} \geq c_0. \tag{5.2}$$

Define the weak curl $\operatorname{curl}_h : \boldsymbol{V}_h \to \boldsymbol{U}_h$ and the discrete Laplacian $L_h : \Sigma_h \to \Sigma_h$ by

$$\begin{aligned}(\operatorname{curl}_h \boldsymbol{F}_h, \boldsymbol{\lambda}_h) &= (\boldsymbol{F}_h, \operatorname{curl} \boldsymbol{\lambda}_h), \quad \boldsymbol{\lambda}_h \in \boldsymbol{U}_h,\\ (L_h\psi_h, \gamma_h) &= (\nabla\psi_h, \nabla\gamma_h), \quad \gamma_h \in \Sigma_h.\end{aligned} \tag{5.3}$$

The discrete product space is

$$\mathbb{V}_h = \boldsymbol{S}_h \times (\Sigma_h)^3 \times \boldsymbol{V}_h \times \Sigma_h. \tag{5.4}$$

For any $\boldsymbol{y}_h = (\boldsymbol{v}_h, \boldsymbol{s}_h, \boldsymbol{F}_h, \psi_h)^\intercal \in \mathbb{V}_h$, its norm is

$$\|\boldsymbol{y}_h\|_{\mathbb{V}_h}^2 = \|\nabla \boldsymbol{v}_h\|^2 + \|\nabla \boldsymbol{s}_h\|^2 + \|\boldsymbol{F}_h\|^2 + \|\operatorname{div} \boldsymbol{F}_h\|^2 + \|\operatorname{curl}_h \boldsymbol{F}_h\|^2 + \|\nabla\psi_h\|^2 + \|L_h\psi_h\|^2. \tag{5.5}$$

In general $\mathbb{V}_h \not\subset \mathbb{V}$: Raviart–Thomas fields need not belong to $H(\operatorname{curl})$, and continuous piecewise linear or quadratic potentials need not have a global L^2 Laplacian. The weak operators are therefore essential, not merely notational replacements.

Choose the initial velocity in the discrete divergence-free space, for example by its L^2 projection onto that space. Choose stated finite element approximations $\boldsymbol{\omega}_h^0$ and $\boldsymbol{m}_h^0$; their L^2 projections onto the indicated spaces are well defined. Determine the initial potential $\varphi_h \in \Sigma_h$ by

$$\mu_0(\nabla\varphi_h^0, \nabla\psi_h) = -(\mu_0 \boldsymbol{m}_h^0 + \boldsymbol{H}_e(0), \nabla\psi_h), \qquad \psi_h \in \Sigma_h. \tag{5.6}$$

5.2 The discrete forms and schemes

The bilinear forms $\mathtt{D}$, $\mathtt{J}$, and $\mathtt{C}$ remain unchanged. Define $\mathtt{A}_h$ from (3.4) by replacing the magnetization curls by curl_h and the potential Laplacians by $-L_h$. Define $\mathtt{B}_h$ from (3.6) with the same replacements.

Lemma 5.1 (Discrete structure)**.** *For $\boldsymbol{x}_h, \boldsymbol{y}_h, \boldsymbol{z}_h \in \mathbb{V}_h$,*

$$\mathtt{A}_h(\boldsymbol{y}_h, \boldsymbol{y}_h) \simeq \|\boldsymbol{y}_h\|_{\mathbb{V}_h}^2, \qquad \mathtt{B}_h(\boldsymbol{z}_h, \boldsymbol{x}_h, \boldsymbol{y}_h) = -\mathtt{B}_h(\boldsymbol{z}_h, \boldsymbol{y}_h, \boldsymbol{x}_h). \tag{5.7}$$

The constants in the energy-norm equivalence depend on the physical parameters, not on h. For fixed h, $\mathtt{B}_h$ is a bounded trilinear form.

Proof. Replace the curls and Laplacians in (3.9) by their weak counterparts. This gives the norm equivalence directly. Skew-symmetry follows from the paired terms and does not require integration by parts or an inverse estimate. Boundedness at a fixed mesh follows from finite dimensionality. □

For a fix integer $l \geq 1$, define

$$\mathfrak{K}_h = \{\boldsymbol{y}_h \in \mathbb{V}_h : \mathtt{C}(\boldsymbol{y}_h, q_h) = 0 \text{ for all } q_h \in Q_h\}. \tag{5.8}$$

The fully discrete p-order temporal scheme reads as: Given $\boldsymbol{x}_h^0, \ldots, \boldsymbol{x}_h^{p-1} \in \mathfrak{K}_h$, use the coefficients and extrapolation weights from Section 4:

$$\widehat{\boldsymbol{x}}_{h,n,i} = \sum_{j=0}^{p-1} \ell_j(c_i) \boldsymbol{x}_h^{n-j}. \tag{5.9}$$

For $n = p-1, \ldots, N-1$, find $\boldsymbol{x}_{h,n,i}, \mathcal{K}_{h,n,i} \in \mathbb{V}_h$ and $P_{h,n,i} \in Q_h$ such that

$$\boldsymbol{x}_{h,n,i} = \boldsymbol{x}_h^n + \Delta t \sum_{j=1}^{s} a_{ij} \mathcal{K}_{h,n,j}, \tag{5.10a}$$

$$\begin{aligned}\mathtt{D}(\mathcal{K}_{h,n,i}, \boldsymbol{y}_h) + \mathtt{A}_h(\boldsymbol{x}_{h,n,i}, \boldsymbol{y}_h) + \mathtt{J}(\boldsymbol{x}_{h,n,i}, \boldsymbol{y}_h) + \mathtt{B}_h(\widehat{\boldsymbol{x}}_{h,n,i}, \boldsymbol{x}_{h,n,i}, \boldsymbol{y}_h)\\ + \mathtt{C}(\boldsymbol{y}_h, P_{h,n,i}) = \langle \mathtt{f}(t_{n,i}), \boldsymbol{y}_h \rangle, \qquad \forall\ \boldsymbol{y}_h \in \mathbb{V}_h,\end{aligned} \tag{5.10b}$$

$$\mathtt{C}(\boldsymbol{x}_{h,n,i}, q_h) = 0, \qquad \forall\ q_h \in Q_h, \qquad i = 1, \ldots, s, \tag{5.10c}$$

$$\boldsymbol{x}_h^{n+1} = \boldsymbol{x}_h^n + \Delta t \sum_{i=1}^{s} b_i \mathcal{K}_{h,n,i}. \tag{5.10d}$$

Theorem 5.2 (Fully discrete solvability and stability)**.** *Assume* (5.2) *and the discrete structural properties. Under* (4.3)*, the higher-order system* (5.10) *has a unique solution for every* $\Delta t > 0$ *and every prescribed predictor in* $\mathbb{V}_h$*. It preserves* $\mathfrak{K}_h$ *and satisfies*

$$E_h(\boldsymbol{x}_h^{n+1}) - E_h(\boldsymbol{x}_h^n) + \Delta t \sum_{i=1}^{s} b_i \|\boldsymbol{x}_{h,n,i}\|_{\mathtt{A}_h}^2 + \frac{\Delta t^2}{2} \sum_{i,j=1}^{s} g_{ij} \mathtt{D}(\mathcal{K}_{h,n,i}, \mathcal{K}_{h,n,j}) = \Delta t \sum_{i=1}^{s} b_i \left\langle \mathtt{f}(t_{n,i}), \boldsymbol{x}_{h,n,i} \right\rangle. \tag{5.11}$$

In particular,

$$E_h(\boldsymbol{x}_h^{n+1}) + \frac{\Delta t}{2} \sum_{i=1}^{s} b_i \|\boldsymbol{x}_{h,n,i}\|_{\mathtt{A}_h}^2 \le E_h(\boldsymbol{x}_h^n) + \frac{\Delta t}{2} \sum_{i=1}^{s} b_i \|\mathtt{f}(t_{n,i})\|_{\mathtt{A}_h^*}^2. \tag{5.12}$$

Proof. On $\mathfrak{K}_h^s$, use the weighted form (4.7) with $\mathtt{A}_h$ and $\mathtt{B}_h$. The same matrix identity (4.8) gives coercivity, and the discrete inf-sup condition recovers the pressures. The constraints also hold for all stage derivatives and for the endpoint update. The proof of (4.9) uses only the stage relations and paired cancellations, so it gives (5.11) directly. Young's inequality gives (5.12). □

For the fully discrete p-order temporal scheme (5.9), one option for computing the initial values $\boldsymbol{x}_h^n$, $1 \le n \le p-1$, is to use p-step linear method [23]. That is, for $0 \le n \le p-1$, set $\boldsymbol{x}_{h,n,i}^{[0]} = \boldsymbol{x}_h^n$. For $\ell = 1, \dots, p$, solve

$$\begin{aligned}
\boldsymbol{x}_{h,n,i}^{[\ell]} &= \boldsymbol{x}_h^n + \Delta t \sum_{j=1}^{s} a_{ij} \mathcal{K}_{h,n,j}^{[\ell]}, \\
\mathtt{D}(\mathcal{K}_{h,n,i}^{[\ell]}, \boldsymbol{y}_h) + \mathtt{A}_h(\boldsymbol{x}_{h,n,i}^{[\ell]}, \boldsymbol{y}_h) + \mathtt{J}(\boldsymbol{x}_{h,n,i}^{[\ell]}, \boldsymbol{y}_h) & \\
\quad + \mathtt{B}_h(\boldsymbol{x}_{h,n,i}^{[\ell-1]}, \boldsymbol{x}_{h,n,i}^{[\ell]}, \boldsymbol{y}_h) + \mathtt{C}(\boldsymbol{y}_h, P_{h,n,i}^{[\ell]}) &= \left\langle \mathtt{f}(t_{n,i}), \boldsymbol{y}_h \right\rangle, \qquad \forall\ \boldsymbol{y}_h \in \mathbb{V}_h, \\
\mathtt{C}(\boldsymbol{x}_{h,n,i}^{[\ell]}, q_h) &= 0, \qquad \forall\ q_h \in Q_h, \qquad i = 1, \dots, s.
\end{aligned} \tag{5.13}$$

We then accept the update

$$\boldsymbol{x}_h^{n+1} = \boldsymbol{x}_h^n + \Delta t \sum_{i=1}^{s} b_i \mathcal{K}_{h,n,i}^{[p]}. \tag{5.14}$$

This involves exactly p linear corrections, rather than a nonlinear solve with an unspecified stopping tolerance.

5.3 Implementation of the fully discrete schemes

The fully discrete pth-order scheme introduced in (5.9) involves the discrete adjoint operators curl_h and L_h. For its implementation, these operators are represented through auxiliary variables. At each Runge–Kutta stage, we write

$$\begin{aligned}
\mathcal{K}_{h,n,i} &= (\mathcal{K}_{h,n,i}^{u}, \mathcal{K}_{h,n,i}^{\omega}, \mathcal{K}_{h,n,i}^{m}, \mathcal{K}_{h,n,i}^{\varphi})^{\mathsf{T}}, & \widehat{\boldsymbol{x}}_{h,n,i} &= (\widehat{\boldsymbol{u}}_{h,n,i}, \widehat{\boldsymbol{\omega}}_{h,n,i}, \widehat{\boldsymbol{m}}_{h,n,i}, \widehat{\varphi}_{h,n,i})^{\mathsf{T}}, \\
\boldsymbol{x}_{h,n,i} &= (\boldsymbol{u}_{h,n,i}, \boldsymbol{\omega}_{h,n,i}, \boldsymbol{m}_{h,n,i}, \varphi_{h,n,i})^{\mathsf{T}}, & \boldsymbol{y}_h &= (\boldsymbol{v}_h, \boldsymbol{s}_h, \boldsymbol{F}_h, \psi_h)^{\mathsf{T}}.
\end{aligned}$$

Equations (5.10b) and (5.10c) can then be written equivalently in the following componentwise form.

(1) Velocity and incompressibility equations.

$$\begin{aligned}
\rho(\mathcal{K}_{h,n,i}^{u}, \boldsymbol{v}_h) + \rho b(\widehat{\boldsymbol{u}}_{h,n,i}, \boldsymbol{u}_{h,n,i}, \boldsymbol{v}_h) + \eta(\nabla \boldsymbol{u}_{h,n,i}, \nabla \boldsymbol{v}_h) + \zeta(\operatorname{curl} \boldsymbol{u}_{h,n,i}, \operatorname{curl} \boldsymbol{v}_h) & \\
- 2\zeta(\boldsymbol{\omega}_{h,n,i}, \operatorname{curl} \boldsymbol{v}_h) - (P_{h,n,i}, \operatorname{div} \boldsymbol{v}_h) + \frac{\mu_0}{2}(\vartheta_{h,n,i}, \widehat{\boldsymbol{m}}_{h,n,i} \cdot \boldsymbol{v}_h) & \\
+ \frac{\mu_0}{2}(\nabla \varphi_{h,n,i} \operatorname{div} \widehat{\boldsymbol{m}}_{h,n,i}, \boldsymbol{v}_h) - \frac{\mu_0}{2}(\boldsymbol{v}_h \times \widehat{\boldsymbol{k}}_{h,n,i}, \nabla \varphi_{h,n,i}) & \\
- \frac{\mu_0}{2}(\widehat{\boldsymbol{m}}_{h,n,i} \times \operatorname{curl} \boldsymbol{v}_h, \nabla \varphi_{h,n,i}) = 0, &
\end{aligned} \tag{5.15a}$$

$$(\operatorname{div} \boldsymbol{u}_{h,n,i}, q_h) = 0. \tag{5.15b}$$

(2) Angular-velocity equation.

$$\begin{aligned}\rho\kappa(\mathcal{K}^{\omega}_{h,n,i},\boldsymbol{s}_h)+\eta'(\nabla\boldsymbol{\omega}_{h,n,i},\nabla\boldsymbol{s}_h)+(\eta'+\lambda')(\operatorname{div}\boldsymbol{\omega}_{h,n,i},\operatorname{div}\boldsymbol{s}_h)+4\zeta(\boldsymbol{\omega}_{h,n,i},\boldsymbol{s}_h)\\+\rho\kappa b(\widehat{\boldsymbol{u}}_{h,n,i},\boldsymbol{\omega}_{h,n,i},\boldsymbol{s}_h)-\mu_0(\widehat{\boldsymbol{m}}_{h,n,i}\times\nabla\varphi_{h,n,i},\boldsymbol{s}_h)\\-2\zeta(\operatorname{curl}\boldsymbol{u}_{h,n,i},\boldsymbol{s}_h)=0.\end{aligned}\tag{5.16}$$

(3) Magnetization equation.

$$\begin{aligned}(\mathcal{K}^{m}_{h,n,i},\boldsymbol{F}_h)+\frac{1}{\tau}(\boldsymbol{m}_{h,n,i},\boldsymbol{F}_h)+\sigma(\operatorname{curl}\boldsymbol{k}_{h,n,i},\boldsymbol{F}_h)+\sigma(\operatorname{div}\boldsymbol{m}_{h,n,i},\operatorname{div}\boldsymbol{F}_h)\\-c(\widehat{\boldsymbol{u}}_{h,n,i},\boldsymbol{m}_{h,n,i},\boldsymbol{F}_h)-\frac{1}{2}(\boldsymbol{m}_{h,n,i}\times\operatorname{curl}\widehat{\boldsymbol{u}}_{h,n,i},\boldsymbol{F}_h)\\-\frac{1}{2}(\widehat{\boldsymbol{u}}_{h,n,i}\times\boldsymbol{k}_{h,n,i},\boldsymbol{F}_h)-\frac{1}{2}(\operatorname{curl}\boldsymbol{z}_{h,n,i},\boldsymbol{F}_h)\\-(\widehat{\boldsymbol{\omega}}_{h,n,i}\times\boldsymbol{m}_{h,n,i},\boldsymbol{F}_h)-\frac{\chi_0}{\tau}(\nabla\varphi_{h,n,i},\boldsymbol{F}_h)=0.\end{aligned}\tag{5.17}$$

(4) Magnetostatic potential equation.

$$\begin{aligned}\mu_0(\mathcal{K}^{\varphi}_{h,n,i},\nabla\psi_h)+\frac{\mu_0(1+\chi_0)+\chi_0}{\tau}(\nabla\varphi_{h,n,i},\nabla\psi_h)+\sigma\mu_0(\nabla\vartheta_{h,n,i},\nabla\psi_h)\\+\mu_0(\boldsymbol{\omega}_{h,n,i}\times\widehat{\boldsymbol{m}}_{h,n,i},\nabla\psi_h)+\frac{\chi_0}{\tau}(\boldsymbol{m}_{h,n,i},\nabla\psi_h)\\-\frac{\mu_0}{2}(\nabla\varpi_{h,n,i},\nabla\psi_h)-\frac{\mu_0}{2}(\boldsymbol{u}_{h,n,i}\operatorname{div}\widehat{\boldsymbol{m}}_{h,n,i},\nabla\psi_h)\\+\frac{\mu_0}{2}(\widehat{\boldsymbol{m}}_{h,n,i}\times\operatorname{curl}\boldsymbol{u}_{h,n,i},\nabla\psi_h)+\frac{\mu_0}{2}(\boldsymbol{u}_{h,n,i}\times\widehat{\boldsymbol{k}}_{h,n,i},\nabla\psi_h)\\+(\partial_t\boldsymbol{H}_e(t_{n,i}),\nabla\psi_h)+\frac{\mu_0+\chi_0}{\tau\mu_0}(\boldsymbol{H}_e(t_{n,i}),\nabla\psi_h)\\+\sigma(\nabla\operatorname{div}\boldsymbol{H}_e(t_{n,i}),\nabla\psi_h)=0.\end{aligned}\tag{5.18}$$

(5) Auxiliary-variable relations. The auxiliary variables

$$\widehat{\boldsymbol{k}}_{h,n,i},\boldsymbol{k}_{h,n,i},\boldsymbol{z}_{h,n,i}\in\boldsymbol{U}_h,\qquad\vartheta_{h,n,i},\varpi_{h,n,i}\in\Sigma_h$$

are defined by

$$\begin{aligned}(\widehat{\boldsymbol{k}}_{h,n,i},\boldsymbol{\theta}_h)&=(\widehat{\boldsymbol{m}}_{h,n,i},\operatorname{curl}\boldsymbol{\theta}_h),&&\forall\,\boldsymbol{\theta}_h\in\boldsymbol{U}_h,\\(\boldsymbol{k}_{h,n,i},\boldsymbol{\theta}_h)&=(\boldsymbol{m}_{h,n,i},\operatorname{curl}\boldsymbol{\theta}_h),&&\forall\,\boldsymbol{\theta}_h\in\boldsymbol{U}_h,\\(\vartheta_{h,n,i},\varsigma_h)&=(\nabla\varphi_{h,n,i},\nabla\varsigma_h),&&\forall\,\varsigma_h\in\Sigma_h,\\(\boldsymbol{z}_{h,n,i},\boldsymbol{\lambda}_h)&=(\widehat{\boldsymbol{u}}_{h,n,i}\times\boldsymbol{m}_{h,n,i},\boldsymbol{\lambda}_h),&&\forall\,\boldsymbol{\lambda}_h\in\boldsymbol{U}_h,\\(\varpi_{h,n,i},\gamma_h)&=(\boldsymbol{u}_{h,n,i}\cdot\widehat{\boldsymbol{m}}_{h,n,i},\gamma_h),&&\forall\,\gamma_h\in\Sigma_h.\end{aligned}\tag{5.19}$$

Remark 5.3. At each endpoint and each Runge–Kutta stage, define

$$\boldsymbol{H}^n_h=\nabla\varphi^n_h\in\boldsymbol{U}_h,\qquad\boldsymbol{H}_{h,n,i}=\nabla\varphi_{h,n,i}\in\boldsymbol{U}_h.\tag{5.20}$$

It is easy to see that

$$\operatorname{curl}\boldsymbol{H}^n_h=0,\qquad\operatorname{curl}\boldsymbol{H}_{h,n,j}=0.\tag{5.21}$$

5.4 An exact block elimination

For a one-stage solve, group the coefficient vectors as

$$\begin{aligned}&X_1=(\boldsymbol{u}_{h,n,i},P_{h,n,i}),&&X_2=\boldsymbol{\omega}_{h,n,i},&&X_3=(\boldsymbol{m}_{h,n,i},\boldsymbol{z}_{h,n,i},\boldsymbol{k}_{h,n,i}),\\&X_4=(\varphi_{h,n,i},\vartheta_{h,n,i}),&&X_5=\varpi_{h,n,i}.\end{aligned}$$

Eliminate the stage derivatives with $\mathsf{S} = \mathsf{A}^{-1}$, after incorporating the known terms into the loads and fixing each stage pressure mean, both the one-stage and multistage systems have the block pattern

$$\begin{pmatrix} K_1 & C_{12} & 0 & C_{14} & 0 \\ C_{21} & K_2 & 0 & C_{24} & 0 \\ 0 & 0 & K_3 & C_{34} & 0 \\ C_{41} & C_{42} & C_{43} & K_4 & C_{45} \\ C_{51} & 0 & 0 & 0 & K_5 \end{pmatrix} \begin{pmatrix} X_1 \\ X_2 \\ X_3 \\ X_4 \\ X_5 \end{pmatrix} = \begin{pmatrix} F_1 \\ F_2 \\ F_3 \\ F_4 \\ F_5 \end{pmatrix}. \tag{5.22}$$

To give a compact exact reduction, set

$$\begin{aligned} H_{12} &= \begin{pmatrix} K_1 & C_{12} \\ C_{21} & K_2 \end{pmatrix}, \quad B_{12} = \begin{pmatrix} C_{14} \\ C_{24} \end{pmatrix}, & F_{12} &= \begin{pmatrix} F_1 \\ F_2 \end{pmatrix}, \\ E_1 &= \begin{pmatrix} I & 0 \end{pmatrix}, & G_{12} &= \begin{pmatrix} C_{41} & C_{42} \end{pmatrix} - C_{45} K_5^{-1} C_{51} E_1. \end{aligned} \tag{5.23}$$

First solve the potential Schur system

$$\begin{aligned} S_4 X_4 &= g_4, \\ S_4 &= K_4 - G_{12} H_{12}^{-1} B_{12} - C_{43} K_3^{-1} C_{34}, \\ g_4 &= F_4 - C_{45} K_5^{-1} F_5 - G_{12} H_{12}^{-1} F_{12} - C_{43} K_3^{-1} F_3. \end{aligned} \tag{5.24}$$

Then recover the other blocks by

$$\begin{pmatrix} X_1 \\ X_2 \end{pmatrix} = H_{12}^{-1}(F_{12} - B_{12} X_4), \qquad X_3 = K_3^{-1}(F_3 - C_{34} X_4), \qquad X_5 = K_5^{-1}(F_5 - C_{51} X_1). \tag{5.25}$$

6 Numerical experiments

In this section, we test convergence and energy decay for the first- through fourth-order schemes, and use the second-order scheme for two magnetically driven channel-flow examples. All computations are performed with FEniCSx/DOLFINx 0.10.0 [5], PETSc, and MUMPS.

For the first-order scheme (4.30), we use the MINI element for the velocity–pressure pair $(\boldsymbol{u}, \tilde{p})$, RT_0 for $\boldsymbol{m}$, NE_0 for $\boldsymbol{H}$, $\boldsymbol{k}$, and $\boldsymbol{z}$, continuous piecewise $\mathbb{P}_1$ vector elements for $\boldsymbol{\omega}$, and continuous piecewise $\mathbb{P}_1$ scalar elements for φ, ϑ, and ϖ in (5.15)–(5.19). For the second-, third-, and fourth-order schemes (4.37), (4.45), and (4.49), respectively, we use the Taylor–Hood pair $\mathbb{P}_k \times \mathbb{P}_{k-1}$ for $(\boldsymbol{u}, \tilde{p})$, RT_{k-1} for $\boldsymbol{m}$, and NE_{k-1} for $\boldsymbol{H}$, $\boldsymbol{k}$, and $\boldsymbol{z}$, where $k = 2, 3, 4$. The variables $\boldsymbol{\omega}$ and φ, ϑ, and ϖ are approximated by continuous piecewise $\mathbb{P}_k$ vector and scalar elements, respectively.

6.1 Convergence test

We compare the first- through fourth-order schemes using one common continuous manufactured solution on $\Omega = (0,1)^3$, with $T = 2$ and all physical parameters $\rho, \kappa, \eta, \zeta, \mu_0, \sigma, \eta', \lambda', \tau, \chi_0$ equal to one. Let

$$a(t) = e^{t/2}, \qquad B(x,y,z) = x(1-x)y(1-y)z(1-z).$$

The prescribed fields are

$$\begin{aligned}
\boldsymbol{u} &= \frac{a(t)}{C_u}\nabla B \times (1,2,3)^{\intercal},\\
\boldsymbol{\omega} &= \frac{a(t)}{C_\omega}(1+x^5+yz^4,\ 2+y^5+xz^4,\ -1+z^5+xy^4)^{\intercal},\\
\boldsymbol{m} &= \frac{a(t)}{C_m}(1+x^5+y^2z^3,\ 2+y^5+z^2x^3,\ 3+z^5+x^2y^3)^{\intercal},\\
\varphi &= \frac{a(t)}{C_H}(1+x^5+2y^5+3z^5+x^2y^2z),\\
\widetilde{p} &= \frac{a(t)}{C_p}(x^4+2y^4-3z^4+xy^2z-1/12), \qquad \boldsymbol{H} = \nabla\varphi, \qquad \boldsymbol{k} = \operatorname{curl}\boldsymbol{m}.
\end{aligned}$$

Here $\widetilde{p}$ is the zero-mean shifted pressure in (3.12); a corresponding physical pressure is $p = \widetilde{p} + \mu_0\boldsymbol{m}\cdot\boldsymbol{H}/2$. The normalization constants, computed from continuous norms at $t = 0$ and fixed for all meshes and orders, are

$$\begin{aligned}
C_u &= 0.5849976258261415, \quad C_\omega = 4.301498111459077,\\
C_m &= 5.636941389185116, \quad C_H = 6.437390775772433,\\
C_p &= 1.0210677573400138.
\end{aligned}$$

The remaining auxiliary fields are $\boldsymbol{z} = \boldsymbol{u}\times\boldsymbol{m}$, $\vartheta = -\operatorname{div}\boldsymbol{H}$, and $\varpi = \boldsymbol{u}\cdot\boldsymbol{m}$.

For this manufactured test, continuous body sources $\boldsymbol{f}$, $\boldsymbol{g}$, and $\boldsymbol{r}$ are added to the momentum, angular-momentum, and magnetization equations, respectively, and are obtained by substitution of the prescribed fields into the strong equations. We take

$$\boldsymbol{H}_e = -\mu_0(\boldsymbol{m}+\boldsymbol{H}), \qquad \boldsymbol{G}_e = -\partial_t\boldsymbol{H}_e - \beta\boldsymbol{H}_e + \sigma\nabla\operatorname{div}\boldsymbol{H}_e.$$

The corresponding load in the potential equation is formed consistently from $\boldsymbol{G}_e - \mu_0\boldsymbol{r}$. The sources and boundary traces are independent of the mesh, the scheme order, the time step, and the computed solution. The particular choice of $\boldsymbol{H}_e$ is part of this manufactured test.

The prescribed traces are generally nonhomogeneous. They are imposed using boundary-data lifts for the four dynamic fields $(\boldsymbol{u},\boldsymbol{\omega},\boldsymbol{m},\varphi)$, including the known lift derivatives and endpoint corrections; the nonlinear predictors retain the numerical physical-field history. Initially, the velocity is projected with its discrete divergence constraint, the angular velocity uses an L^2 projection, and the magnetization uses an L^2 projection constrained by the prescribed normal trace and the moments of its divergence in the corresponding discontinuous polynomial space. The scalar potential is initialized by the compatible Maxwell–Poisson constraint with the prescribed trace. The auxiliary endpoint fields $\boldsymbol{k}$ and ϑ are recovered through their weak mass relations. These choices specify the nonhomogeneous extension used in this experiment; the homogeneous boundary assumptions of the preceding analysis are not silently imposed on this manufactured solution. The energy-decay experiment below has its own homogeneous-boundary initial data.

We use $K\times K\times K$ uniform tetrahedral meshes and $\Delta t = 1/(2K)$. The first- and second-order schemes use $K = 2,4,8,16,32$, the third-order scheme uses $K = 2,4,8,16$, and the fourth-order scheme uses $K = 2,4,8$. Thus the comparison contains seventeen completed cases. The finite element spaces are those specified above, and the starting steps use the prescribed fixed number of linear corrections. Space and time are refined together in this test.

For $v\in\{\boldsymbol{u},\boldsymbol{\omega},\boldsymbol{m},\boldsymbol{H},\boldsymbol{k},\varphi\}$, let X_v denote, respectively, H^1, H^1, $H(\operatorname{div})$, $H(\operatorname{curl})$, L^2, and H^1. We report

$$E_{v,\mathrm{abs}} = \|v_h^N - v(T)\|_{X_v}, \qquad E_{v,\mathrm{rel}} = \frac{E_{v,\mathrm{abs}}}{\|v(T)\|_{X_v}}.$$

The H^1, $H(\operatorname{div})$, and $H(\operatorname{curl})$ quantities are full norms, not seminorms. Pressure is measured at the actual Runge–Kutta stages of the last time step. With $t_{N-1,i} = T - \Delta t + c_i\Delta t$ and

$v^\circ = v - |\Omega|^{-1}\int_\Omega v\,\mathrm{d}x$, its errors are

$$E_{\widetilde{p},\mathrm{abs}} = \left(\sum_{i=1}^{s} b_i \|P^\circ_{h,N-1,i} - \widetilde{p}(t_{N-1,i})^\circ\|^2_{L^2}\right)^{1/2},$$
$$R_{\widetilde{p}} = \left(\sum_{i=1}^{s} b_i \|\widetilde{p}(t_{N-1,i})^\circ\|^2_{L^2}\right)^{1/2}, \qquad E_{\widetilde{p},\mathrm{rel}} = E_{\widetilde{p},\mathrm{abs}}/R_{\widetilde{p}}.$$

The table heading $\widetilde{p}_{\mathrm{stg}}$ denotes this stage-weighted quantity. It coincides with an endpoint pressure error for backward Euler, whose only stage has $c_1 = 1$. For the Gauss schemes it is neither an endpoint pressure error at T nor a pressure norm accumulated over all time steps. For either error measure, the observed rate between consecutive meshes is

$$\mathrm{Rate}(K_1, K_2) = \frac{\log(E(K_1)/E(K_2))}{\log(K_2/K_1)}.$$

Rates are evaluated from the unrounded errors. The first row of each table has no preceding mesh and is marked by a dash. The quantity

$$C_{\mathrm{curl}} = \max_{x\in\mathcal{Q}_h}\max_{j=1,2,3} |(\mathrm{curl}\,\boldsymbol{H}_h(x))_j|$$

is an absolute sampled curl diagnostic on the retained elementwise evaluation points $\mathcal{Q}_h$, not a separately proved continuous L^∞ norm; no convergence rate is assigned to it.

Tables 1 to 4 give all relative errors and rates. The corresponding absolute errors and their rates are listed in Appendix A. In particular, the pressure reference norm $R_{\widetilde{p}}$ varies with the stage times, so its absolute and relative rates need not be identical.

Table 1: Relative errors and observed rates for the first-order scheme under joint space–time refinement.

(a) Velocity, stage pressure, angular velocity, and magnetization

K	$\boldsymbol{u}$: H^1		$\widetilde{p}_{\mathrm{stg}}$: L^2		$\boldsymbol{\omega}$: H^1		$\boldsymbol{m}$: $H(\mathrm{div})$	
	Error	Rate	Error	Rate	Error	Rate	Error	Rate
2	9.1998×10^{-1}	—	3.6029×10^{0}	—	4.7983×10^{-1}	—	2.7209×10^{-1}	—
4	4.9902×10^{-1}	0.883	1.9199×10^{0}	0.908	2.5935×10^{-1}	0.888	1.4356×10^{-1}	0.922
8	2.3698×10^{-1}	1.074	6.3988×10^{-1}	1.585	1.3223×10^{-1}	0.972	7.2663×10^{-2}	0.982
16	1.1418×10^{-1}	1.053	1.8834×10^{-1}	1.764	6.6435×10^{-2}	0.993	3.6451×10^{-2}	0.995
32	5.6150×10^{-2}	1.024	5.5699×10^{-2}	1.758	3.3259×10^{-2}	0.998	1.8244×10^{-2}	0.999

(b) Magnetic field, auxiliary curl, scalar potential, and curl diagnostic

K	$\boldsymbol{H}$: $H(\mathrm{curl})$		$\boldsymbol{k}$: L^2		φ: H^1		C_{curl}
	Error	Rate	Error	Rate	Error	Rate	
2	5.8948×10^{-1}	—	5.0453×10^{-1}	—	5.6594×10^{-1}	—	2.2627×10^{-14}
4	3.2345×10^{-1}	0.866	2.6648×10^{-1}	0.921	3.0609×10^{-1}	0.887	2.4675×10^{-13}
8	1.6511×10^{-1}	0.970	1.2951×10^{-1}	1.041	1.5575×10^{-1}	0.975	1.9068×10^{-12}
16	8.2963×10^{-2}	0.993	6.3442×10^{-2}	1.030	7.8197×10^{-2}	0.994	6.6707×10^{-12}
32	4.1531×10^{-2}	0.998	3.1471×10^{-2}	1.011	3.9138×10^{-2}	0.999	2.5781×10^{-11}

Table 2: Relative errors and observed rates for the second-order scheme under joint space–time refinement.

(a) Velocity, stage pressure, angular velocity, and magnetization

K	$\boldsymbol{u}$: H^1		$\widetilde{p}_{\mathrm{stg}}$: L^2		$\boldsymbol{\omega}$: H^1		$\boldsymbol{m}$: $H(\mathrm{div})$	
	Error	Rate	Error	Rate	Error	Rate	Error	Rate
2	2.4292×10^{-1}	—	3.4826×10^{-1}	—	1.1186×10^{-1}	—	5.5757×10^{-2}	—
4	6.3590×10^{-2}	1.934	6.3965×10^{-2}	2.445	2.9695×10^{-2}	1.913	1.4449×10^{-2}	1.948
8	1.5959×10^{-2}	1.994	1.2809×10^{-2}	2.320	7.5209×10^{-3}	1.981	3.6331×10^{-3}	1.992
16	3.9912×10^{-3}	1.999	3.0054×10^{-3}	2.091	1.8854×10^{-3}	1.996	9.0923×10^{-4}	1.998
32	9.9784×10^{-4}	2.000	7.4090×10^{-4}	2.020	4.7162×10^{-4}	1.999	2.2735×10^{-4}	2.000

(b) Magnetic field, auxiliary curl, scalar potential, and curl diagnostic

K	$\boldsymbol{H}$: $H(\mathrm{curl})$		$\boldsymbol{k}$: L^2		φ: H^1		C_{curl}
	Error	Rate	Error	Rate	Error	Rate	
2	1.3863×10^{-1}	—	1.5901×10^{-1}	—	1.3103×10^{-1}	—	5.3910×10^{-13}
4	3.6993×10^{-2}	1.906	4.6644×10^{-2}	1.769	3.4885×10^{-2}	1.909	2.4634×10^{-12}
8	9.4092×10^{-3}	1.975	1.2376×10^{-2}	1.914	8.8681×10^{-3}	1.976	1.1638×10^{-11}
16	2.3631×10^{-3}	1.993	3.1803×10^{-3}	1.960	2.2269×10^{-3}	1.994	4.7585×10^{-11}
32	5.9148×10^{-4}	1.998	8.0496×10^{-4}	1.982	5.5737×10^{-4}	1.998	2.3508×10^{-10}

Table 3: Relative errors and observed rates for the third-order scheme under joint space–time refinement.

(a) Velocity, stage pressure, angular velocity, and magnetization

K	$\boldsymbol{u}$: H^1		$\widetilde{p}_{\mathrm{stg}}$: L^2		$\boldsymbol{\omega}$: H^1		$\boldsymbol{m}$: $H(\mathrm{div})$	
	Error	Rate	Error	Rate	Error	Rate	Error	Rate
2	4.9719×10^{-2}	—	8.7826×10^{-2}	—	1.2586×10^{-2}	—	6.0733×10^{-3}	—
4	6.2086×10^{-3}	3.001	6.3314×10^{-3}	3.794	1.5606×10^{-3}	3.012	7.6424×10^{-4}	2.990
8	7.9045×10^{-4}	2.974	6.1707×10^{-4}	3.359	1.9245×10^{-4}	3.020	9.5462×10^{-5}	3.001
16	1.0057×10^{-4}	2.974	7.3209×10^{-5}	3.075	2.3826×10^{-5}	3.014	1.1922×10^{-5}	3.001

(b) Magnetic field, auxiliary curl, scalar potential, and curl diagnostic

K	$\boldsymbol{H}$: $H(\mathrm{curl})$		$\boldsymbol{k}$: L^2		φ: H^1		C_{curl}
	Error	Rate	Error	Rate	Error	Rate	
2	1.5168×10^{-2}	—	2.6528×10^{-2}	—	1.4314×10^{-2}	—	1.8790×10^{-12}
4	1.8773×10^{-3}	3.014	3.7313×10^{-3}	2.830	1.7696×10^{-3}	3.016	9.6025×10^{-12}
8	2.3103×10^{-4}	3.022	4.8010×10^{-4}	2.958	2.1772×10^{-4}	3.023	4.3286×10^{-11}
16	2.8563×10^{-5}	3.016	6.0418×10^{-5}	2.990	2.6916×10^{-5}	3.016	2.0974×10^{-10}

Table 4: Relative errors and observed rates for the fourth-order scheme under joint space–time refinement.

(a) Velocity, stage pressure, angular velocity, and magnetization

K	$\boldsymbol{u}$: H^1		$\widetilde{p}_{\mathrm{stg}}$: L^2		$\boldsymbol{\omega}$: H^1		$\boldsymbol{m}$: $H(\mathrm{div})$	
	Error	Rate	Error	Rate	Error	Rate	Error	Rate
2	5.5558×10^{-3}	—	6.3611×10^{-3}	—	7.0039×10^{-4}	—	3.5930×10^{-4}	—
4	3.3708×10^{-4}	4.043	2.5658×10^{-4}	4.632	4.1595×10^{-5}	4.074	2.1826×10^{-5}	4.041
8	2.0775×10^{-5}	4.020	1.1129×10^{-5}	4.527	2.5429×10^{-6}	4.032	1.3385×10^{-6}	4.027

(b) Magnetic field, auxiliary curl, scalar potential, and curl diagnostic

K	$\boldsymbol{H}$: $H(\mathrm{curl})$		$\boldsymbol{k}$: L^2		φ: H^1		C_{curl}
	Error	Rate	Error	Rate	Error	Rate	
2	8.3775×10^{-4}	—	4.3619×10^{-3}	—	7.9002×10^{-4}	—	1.3566×10^{-11}
4	5.0372×10^{-5}	4.056	3.0555×10^{-4}	3.835	4.7476×10^{-5}	4.057	3.1267×10^{-11}
8	3.0809×10^{-6}	4.031	1.9963×10^{-5}	3.936	2.9033×10^{-6}	4.031	1.5759×10^{-10}

On the finest adjacent meshes, the six nonpressure norms exhibit rates close to the nominal first, second, third, and fourth orders. Pressure has the measured finest-segment relative rates 1.7576, 2.0202, 3.0754, and 4.5270, respectively. For the fourth-order scheme, the absolute pressure rate is 4.5048; the difference of about 0.0222 comes from the changing stage reference norm. The first-order reference pressure norm is evaluated at T on every mesh, so its absolute and relative rates agree up to roundoff.

These pressure values are finite-grid observations. Contributions from space, time, initialization, and the nonhomogeneous boundary treatment have not been separated by this joint refinement. In particular, Theorem 4.4 estimates the endpoint and accumulated stage errors of the state variables, not this last-step pressure norm. The tables therefore do not establish an independent temporal or spatial order, or a general pressure superconvergence result. The sampled curl values are reported as finite-precision diagnostics of the gradient representation. The third-order $K = 8$ endpoint was independently remeasured from archived terminal fields after a monitoring failure; no time step was repeated for that endpoint measurement.

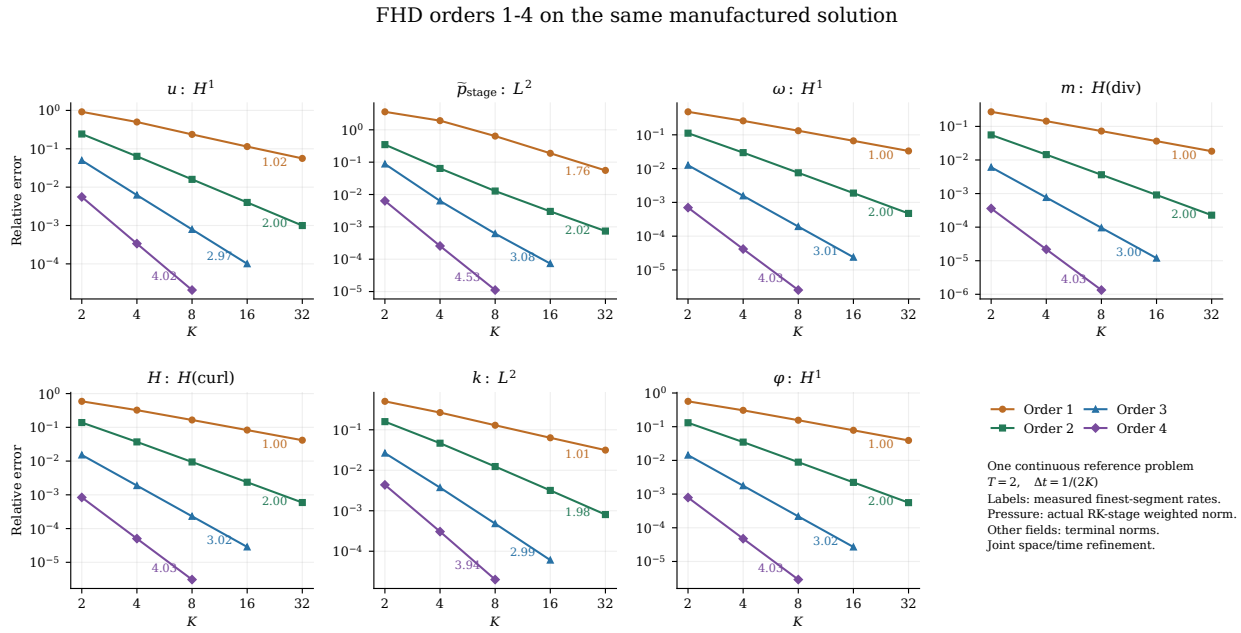


Figure 1: Relative errors for the first- through fourth-order schemes on the common manufactured solution, $T = 2$ and $\Delta t = 1/(2K)$. Each panel uses its stated norm. The numerical labels are the observed rates on the finest adjacent meshes, not prescribed integer slopes.

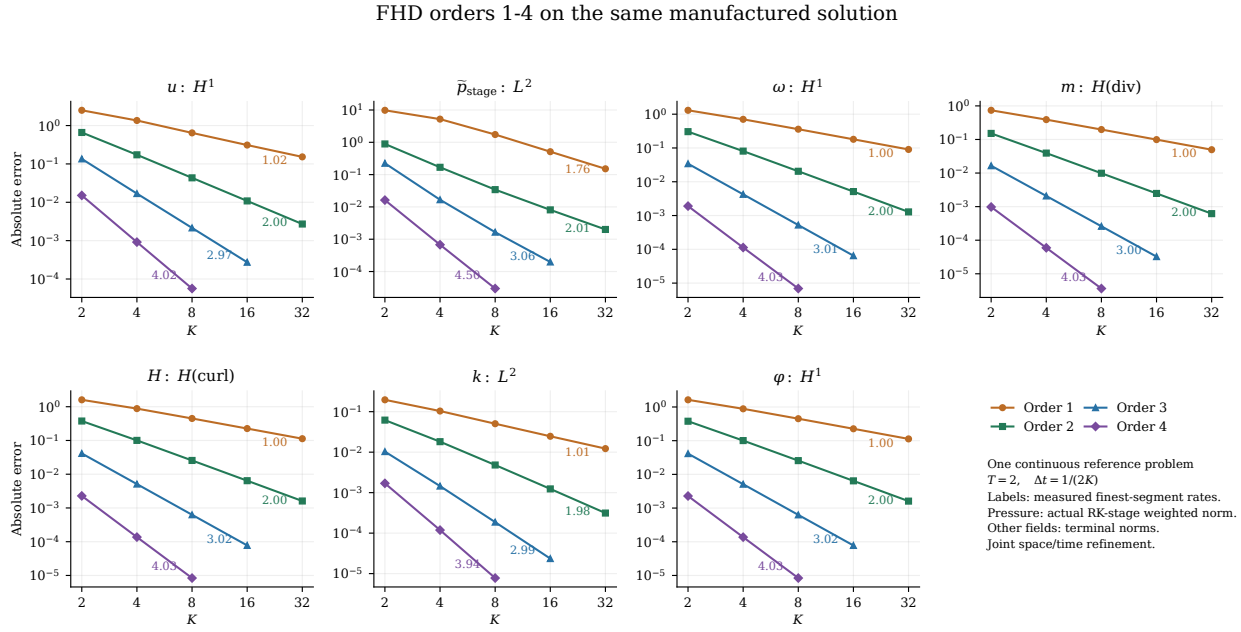


Figure 2: Absolute errors for the same seventeen accepted cases. Pressure uses the actual last-step Runge–Kutta stage-weighted norm; the other fields use terminal norms. Spatial and temporal refinement are coupled in this experiment.

6.2 Energy dissipation test

This subsection is to investigate the energy decaying phenomenon of the kth-order ($k = 1, 2, 3, 4$) temporal schemes (4.30), (4.37), (4.45), and (4.49). The finite element families are chosen as in Section 6.1. We consider the FHD model (2.1) with external magnetic field $\boldsymbol{H}_e = 0$ and homogeneous boundary conditions. The initial value functions are specified below. The symbols in these initial data are defined independently of the manufactured solution in Section 6.1. Specifically,

$$\begin{aligned} q_{\mathrm{E}}(s) &= 64s^3(1-s)^3, \qquad \Phi_{\mathrm{E}}(x,y,z) = q_{\mathrm{E}}(x)q_{\mathrm{E}}(y)q_{\mathrm{E}}(z), \\ S_{\mathrm{E}}(x,y,z) &= \sin^2(\pi x)\sin^2(\pi y)\sin^2(\pi z), \\ \boldsymbol{a}_{\mathrm{E}} &= (1,2,3)^{\mathsf{T}}, \quad \boldsymbol{b}_{\mathrm{E}} = (1,-1,1/2)^{\mathsf{T}}, \quad \boldsymbol{c}_{\mathrm{E}} = (1,2,-1)^{\mathsf{T}}, \\ C_{u,\mathrm{E}} &= 363.275188821, \quad C_{\omega,\mathrm{E}} = 20.6500992827, \quad C_{m,\mathrm{E}} = \frac{\sqrt{81+108\pi^2}}{16}. \end{aligned}$$

Thus the original initial fields used for the energy curves remain

$$\boldsymbol{u}_0 = C_{u,\mathrm{E}}^{-1}\nabla\Phi_{\mathrm{E}} \times \boldsymbol{a}_{\mathrm{E}}, \qquad \boldsymbol{\omega}_0 = C_{\omega,\mathrm{E}}^{-1}\Phi_{\mathrm{E}}\boldsymbol{b}_{\mathrm{E}}, \qquad \boldsymbol{m}_0 = C_{m,\mathrm{E}}^{-1}S_{\mathrm{E}}\boldsymbol{c}_{\mathrm{E}}.$$

All physical parameters are one. The body sources and external magnetic field are zero, and the essential boundary traces are homogeneous. We use $K = 8$, $T = 0.5$, and $\Delta t = 1/16$, $1/32$, $1/64$, giving 8, 16, and 32 time steps, respectively. The plotted quantity is E_h^n/E_h^0, where

$$E_h^n = \frac{1}{2}\left(\rho\|\boldsymbol{u}_h^n\|_0^2 + \rho\kappa\|\boldsymbol{\omega}_h^n\|_0^2 + \|\boldsymbol{m}_h^n\|_0^2 + \mu_0\|\nabla\varphi_h^n\|_0^2\right).$$

The first- and second-order curves are retained from the original energy experiment. The six supplemental third- and fourth-order runs use exactly the continuous initial fields above. Their initial velocity is the constrained L^2 projection onto the discrete divergence-free subspace; $\boldsymbol{\omega}_0$ and $\boldsymbol{m}_0$ are projected in L^2, and the initial magnetic potential satisfies the discrete zero-field Maxwell–Poisson constraint. The third-order method uses two Gauss stages and three linear corrections on its first two steps; the fourth-order method uses three Gauss stages and four corrections on its first three steps. Thereafter, the prescribed history extrapolation is used.

Figure 3 shows nonincreasing normalized energy at all stored time levels. For the six supplemental runs, we also check the zero-load Gauss energy identity (5.11) at every step, including startup. With $D_h^n = \Delta t\sum_i b_i\|\boldsymbol{x}_{h,n,i}\|_{\mathtt{A}_h}^2$, the maximum scaled residual $|E_h^{n+1} - E_h^n + D_h^n|/(E_h^{n+1} + E_h^n + D_h^n)$ is 5.87×10^{-12}. These curves illustrate dissipation for the tested steps. Each temporal order uses its corresponding spatial finite-element spaces, so this comparison does not isolate temporal accuracy.

6.3 Magnetically forced flows

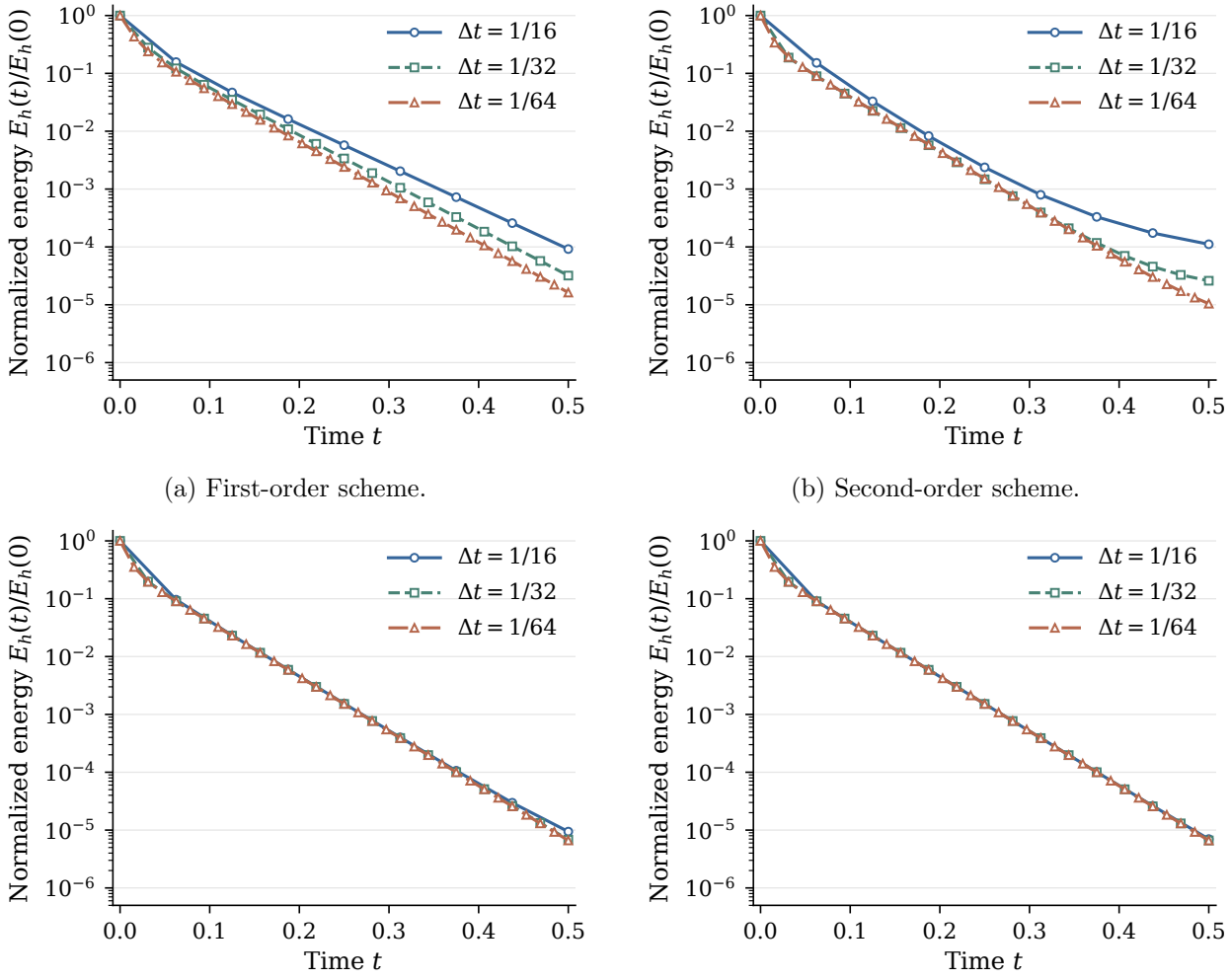
In this subsection, we apply the second-order scheme (4.37) to a three-dimensional ferrofluid transport problem with prescribed inflow. These examples are inspired by the channel-pumping tests in Dong et al. [7], but it does not reproduce their parameter settings. Instead, we retain the magnetic diffusion, edge and face finite element spaces, and midpoint–extrapolation scheme used in the present work. All quantities are dimensionless.

Example 1 (Magnetically forced flow in a straight channel)**.** We consider the straight channel $\Omega = (0,1) \times (0,0.2) \times (0,0.2)$, with inlet $\Gamma_i = \{x = 0\}$, outlet $\Gamma_o = \{x = 1\}$, and solid walls $\Gamma_w = \partial\Omega \setminus (\Gamma_i \cup \Gamma_o)$. The material parameters are

$$\rho = \mu_0 = \chi_0 = 1, \qquad \kappa = \zeta = 0.01, \qquad \eta = 0.02, \qquad \eta' = \lambda' = 0.001, \qquad \tau = 0.1, \qquad \sigma = 0.005,$$

and all body sources and initial values are set to zero. The inlet velocity is

$$\boldsymbol{u}_D = (4R(t/0.4)Y(1-Y)Z(1-Z), 0, 0)^{\mathsf{T}}, \qquad Y = \frac{y}{0.2}, \quad Z = \frac{z}{0.2},$$



(a) First-order scheme.

(b) Second-order scheme.

(c) Third-order scheme.

(d) Fourth-order scheme.

Figure 3: Normalized discrete energy for the first- through fourth-order schemes, with $K = 8$, $T = 0.5$, zero forcing and homogeneous boundary traces. All panels use the same axes; each curve is normalized by its own discrete initial energy.

where

$$R(s)=\begin{cases}0, & s\le 0,\\ 10s^3-15s^4+6s^5, & 0<s<1,\\ 1, & s\ge 1.\end{cases}$$

The magnetic field is generated by eight dipoles located outside the channel at $\boldsymbol{x}_i^- = (0.2i, -0.15, 0.1)$ and $\boldsymbol{x}_i^+ = (0.2i, 0.35, 0.1)$, $i = 1, \dots, 4$, with directions $\boldsymbol{d}_i^- = (0, 1, 0)^\intercal$ and $\boldsymbol{d}_i^+ = (0, -1, 0)^\intercal$. Their potential and time-dependent strengths are

$$\Phi_a(\boldsymbol{x},t)=\sum_{i=1}^{4} a_i(t)\sum_{\epsilon\in\{-,+\}}\frac{\boldsymbol{d}_i^\epsilon\cdot(\boldsymbol{x}_i^\epsilon-\boldsymbol{x})}{|\boldsymbol{x}_i^\epsilon-\boldsymbol{x}|^3}, \qquad \boldsymbol{h}_a=\nabla\Phi_a, \tag{6.1}$$

$$a_i(t)=C_aAR\left(\frac{t-0.2}{0.4}\right)\left[0.7+0.3\cos\left(2\pi\left(t-\frac{0.2i}{0.8}\right)\right)\right], \qquad C_a=3\times10^{-4}, \tag{6.2}$$

with $\boldsymbol{H}_e = -\mu_0\boldsymbol{h}_a$. We take $A = 1$ for the magnetic case and $A = 0$ for the zero-field control. Since the dipoles lie outside Ω, Φ_a is harmonic in the fluid region, and the magnetic-potential load is

$$(\boldsymbol{G}_e,\nabla\psi_h), \qquad \boldsymbol{G}_e=\mu_0\partial_t\boldsymbol{h}_a+\frac{\mu_0+\chi_0}{\tau}\boldsymbol{h}_a.$$

We impose $\boldsymbol{u} = \boldsymbol{0}$ on Γ_w, the gradient-traction condition

$$((\eta+\zeta)\nabla\boldsymbol{u}-pI)\boldsymbol{n}=\boldsymbol{0} \qquad \text{on } \Gamma_o,$$

and

$$\boldsymbol{\omega}=\boldsymbol{0}, \qquad \boldsymbol{m}\cdot\boldsymbol{n}=0, \qquad \boldsymbol{k}\times\boldsymbol{n}=\boldsymbol{0}, \qquad (\boldsymbol{H}+\boldsymbol{m})\cdot\boldsymbol{n}=\boldsymbol{h}_a\cdot\boldsymbol{n} \qquad \text{on } \partial\Omega,$$

together with $\int_\Omega \varphi_h\,\mathrm{d}x = 0$. The prescribed normal magnetic load is included in the weak definition of ϑ_h.

We set $\Delta t = 0.005$ and $T = 2$. The magnetic simulations were performed on meshes with $K = 4, 8, 12$, while the zero-field control used $K = 12$. As shown in Figures 4 to 6, the velocity at $t = 2$ remains mainly aligned with the channel, whereas the magnetization follows the nonuniform dipole field and the particle angular velocity is concentrated in regions of strong shear; in particular, it is different from the fluid vorticity $\operatorname{curl}\boldsymbol{u}_h$. The relative L^2 differences between the $K = 8$ and $K = 12$ solutions are 0.13524%, 3.29718%, 0.71893%, and 3.14611% for $\boldsymbol{u}$, $\boldsymbol{m}$, $\boldsymbol{\omega}$, and $\boldsymbol{H}$, respectively. These values measure differences between two successive meshes and are not errors relative to an exact solution. In the zero-field control, both the magnetization and magnetic field vanish, while the differences from the magnetic case are 0.00725% in velocity and 0.15803% in angular velocity. The results show the coupled response to the applied magnetic field, but they do not by themselves establish mesh-converged magnetic pumping or an increase in the prescribed through-flow.

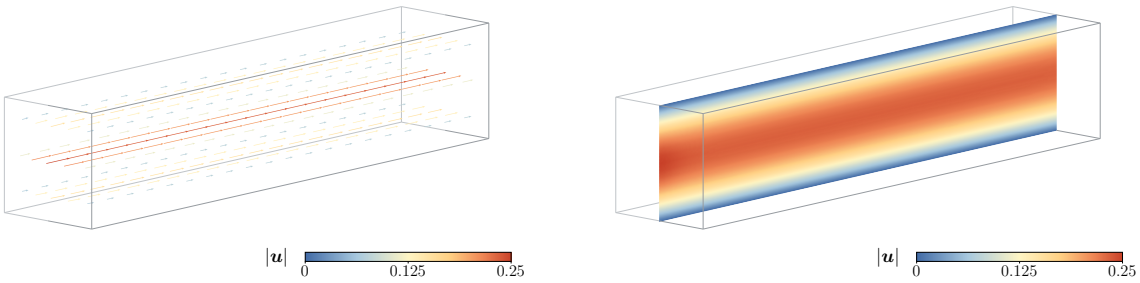


Figure 4: Straight-channel flow at $t = 2$ with $K = 12$: velocity vectors (left) and the velocity magnitude on $z = 0.1$ (right).

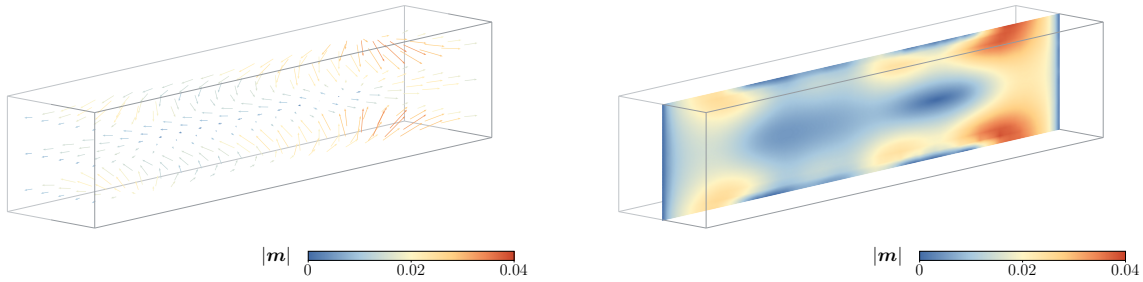


Figure 5: Straight-channel flow at $t = 2$ with $K = 12$: magnetization vectors (left) and the magnetization magnitude on $z = 0.1$ (right).

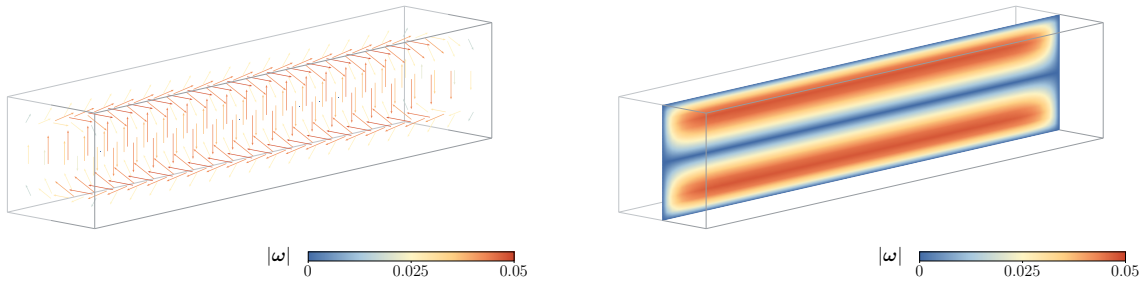


Figure 6: Straight-channel flow at $t = 2$ with $K = 12$: particle angular velocity vectors (left) and their magnitude on $z = 0.1$ (right). The arrows display instantaneous vectors, not particle trajectories.

Example 2 (Magnetically forced flow over a backward-facing step)**.** This example, we consider the backward-facing-step channel

$$\Omega = (0,1) \times (0,0.2) \times (0,0.2) \setminus ([0,0.2] \times [0,0.2] \times [0,0.1]), \tag{6.3}$$

which is motivated by the three-dimensional configuration in [7]. Unless stated otherwise, the material parameters, zero initial data, body forces, finite element spaces, boundary conditions, time step, final time, quadrature rule, and solver tolerances are the same as those used in Example 1. The inlet is the upper half of the face $x = 0$, and the prescribed velocity is

$$\boldsymbol{u}_D = (4R(t/0.4)Y(1-Y)Z_s(1-Z_s), 0, 0)^{\intercal}, \qquad Y = \frac{y}{0.2}, \qquad Z_s = \frac{z-0.1}{0.1}, \tag{6.4}$$

where R is defined in Example 1. The same magnetic potential and time-dependent waveform as in (6.1)–(6.2) are used, but the dipoles are placed at

$$\boldsymbol{x}_i^- = (0.2i, -0.2, 0.1), \qquad \boldsymbol{x}_i^+ = (0.2i, 0.4, 0.1), \qquad i = 1, \ldots, 4,$$

with

$$\boldsymbol{d}_i^- = (0,1,0)^{\intercal}, \qquad \boldsymbol{d}_i^+ = (0,-1,0)^{\intercal}, \qquad C_a = 7 \times 10^{-4}.$$

A conforming tetrahedral mesh is obtained from a $5K \times K \times K$ background partition after removing the solid cells. To refine the region near the step, the axial coordinate is mapped according to

ξ	0	0.1	0.2	0.6	1
x	0	0.15	0.2	0.4	1

.

The resulting mesh contains $27K^3$ tetrahedra. The results below use $K = 4$ and show the solution at the stored time level $t = 0.715$. As illustrated in Figures 7 to 9, the velocity expands into the

larger downstream section, the magnetization follows the nonuniform applied field, and the particle angular velocity is strongest near the step and the channel walls. The maximum relative algebraic residual over the simulation is 2.03×10^{-10}, and the largest relative inlet–outlet flux imbalance is 1.12×10^{-14}. A comparison with the smaller time step $\Delta t = 0.0025$ gives endpoint relative L^2 differences below 2.9×10^{-5} for $\boldsymbol{u}$, $\boldsymbol{m}$, $\boldsymbol{\omega}$, and $\boldsymbol{H}$. These results confirm the accuracy of the algebraic solver and show low sensitivity to the chosen time step. However, the $K = 4$ results are intended only as a qualitative illustration; mesh-independent flow quantities are not claimed. The nonconvex geometry and open boundaries also lie outside the homogeneous convex-domain setting of the analysis.

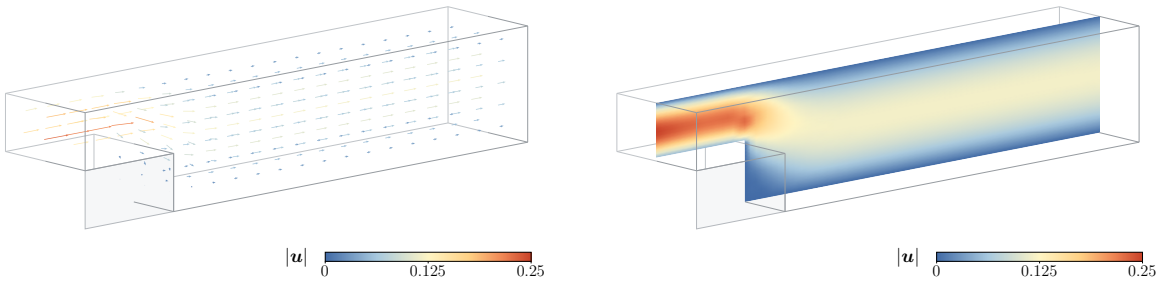


Figure 7: Backward-facing-step flow at $t = 0.715$ with $K = 4$: velocity vectors (left) and velocity magnitude on $y = 0.1$ (right). The unfilled upstream region is the solid step.

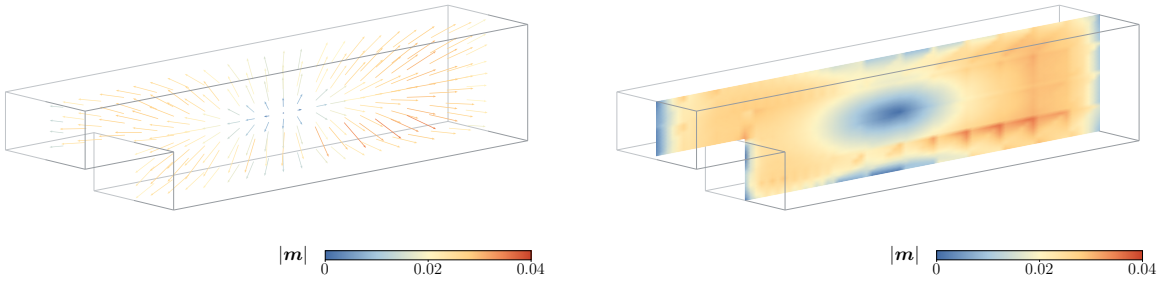


Figure 8: Backward-facing-step flow at $t = 0.715$ with $K = 4$: magnetization vectors (left) and magnetization magnitude on $y = 0.1$ (right).

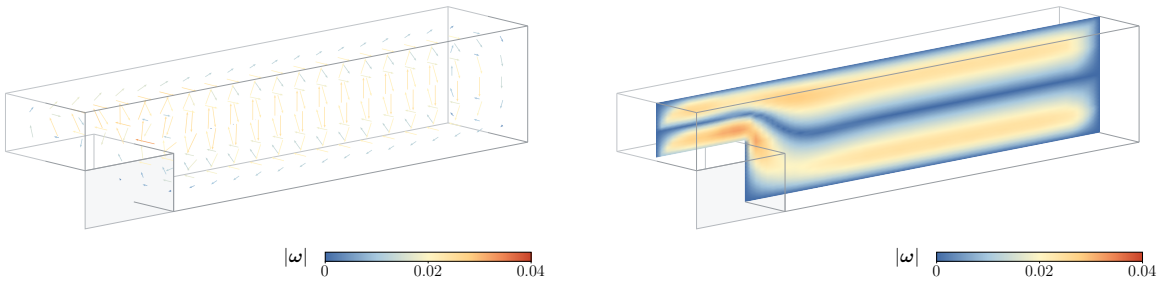


Figure 9: Backward-facing-step flow at $t = 0.715$ with $K = 4$: particle angular-velocity vectors (left) and their magnitude on $y = 0.1$ (right).

6.4 Third-order K8 application snapshots

We additionally display the completed third-order trajectories for the two geometries and physical conditions specified above. Both use $K = 8$, third-order time discretization and the corresponding third-order compatible finite-element spaces, $\Delta t = 0.005$, and $T = 2$. The velocity–pressure pair is $\mathbb{P}_3/\mathbb{P}_2$. The first two time steps use three startup corrections each; the subsequent steps use the retained three-level history. Each case has 400 accepted time steps and 401 saved states, including the zero initial state. All six figures below use the actual endpoint at $t = 2$.

Following the paired-panel presentation of Dong et al. [7], the left panels show sparse instantaneous three-dimensional vectors, and the right panels show the full three-component magnitude on a center section. The sections are $z = 0.1$ for the channel and $y = 0.1$ for the step geometry. The step's solid region is excluded. Arrows are vector glyphs, not integrated streamlines or particle paths; $\boldsymbol{\omega}$ denotes particle angular velocity, not $\operatorname{curl} \boldsymbol{u}$. The paired panels share a linear color scale fixed over the saved trajectory. Colors between section samples are interpolated only for display; the data are not temporally interpolated or polynomially recovered.

The original magnetic diffusion, forcing, and boundary conditions are retained. Thus the common visual presentation does not imply a reproduction of Dong et al.'s different physical model or parameter set. The preceding second-order mesh comparisons remain historical benchmark results. These single-mesh third-order snapshots do not establish a new convergence rate, mesh independence, or a steady or periodic limiting state.

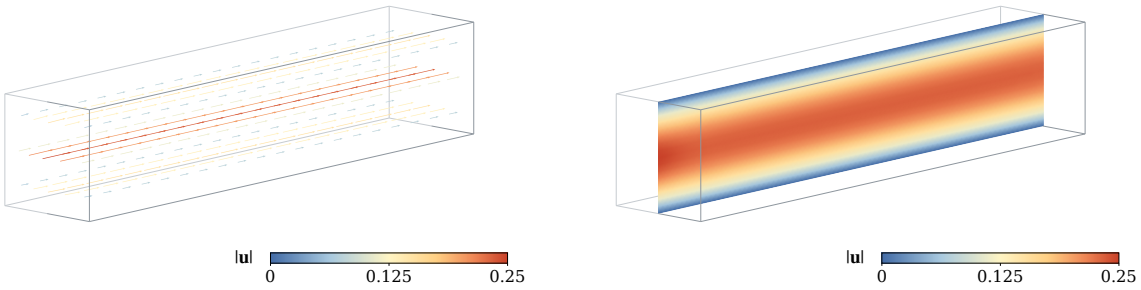


Figure 10: Straight-channel flow, third order in time and space, $K = 8$, $t = 2$: velocity vectors (left) and their magnitude on $z = 0.1$ (right).

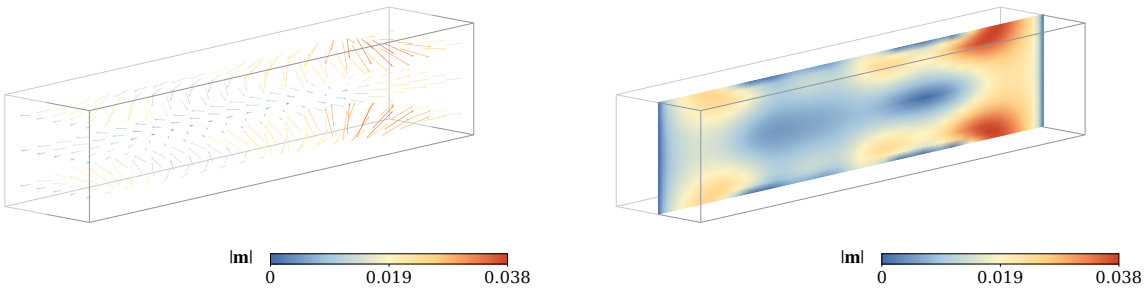


Figure 11: Straight-channel flow, third order in time and space, $K = 8$, $t = 2$: magnetization vectors (left) and their magnitude on $z = 0.1$ (right).

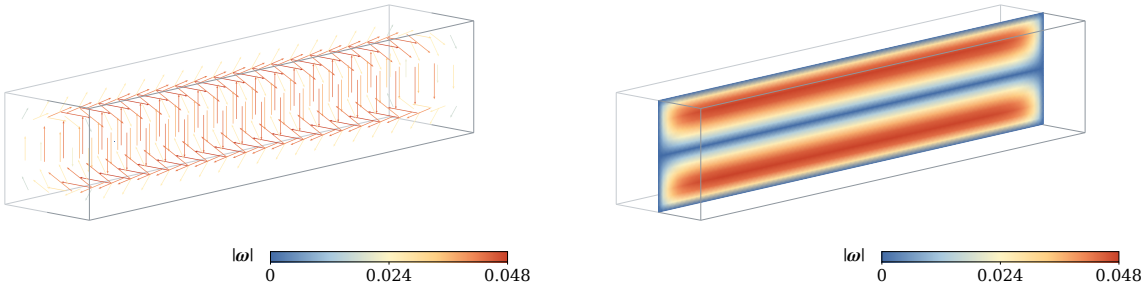


Figure 12: Straight-channel flow, third order in time and space, $K = 8$, $t = 2$: particle angular velocity vectors (left) and their magnitude on $z = 0.1$ (right).

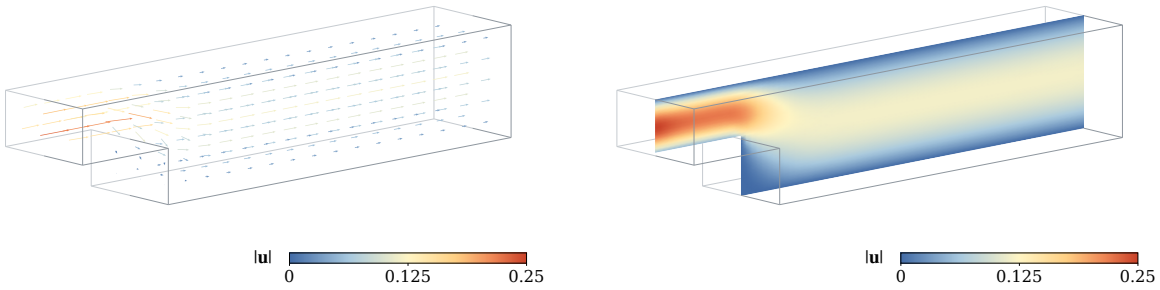


Figure 13: Backward-facing-step flow, third order in time and space, $K = 8$, $t = 2$: velocity vectors (left) and their magnitude on $y = 0.1$ (right).

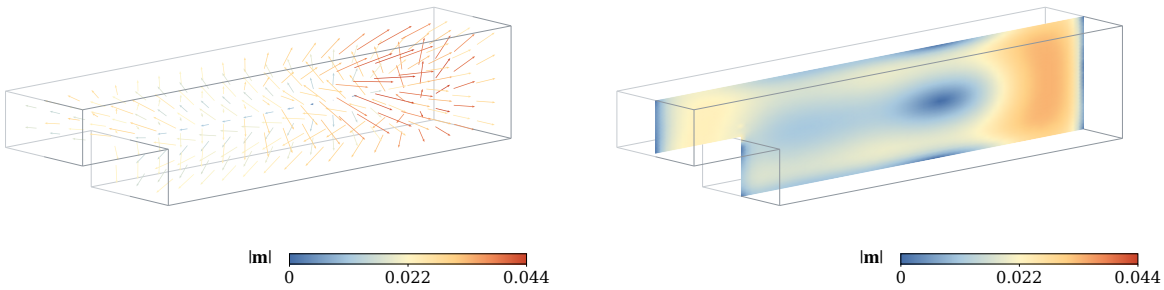


Figure 14: Backward-facing-step flow, third order in time and space, $K = 8$, $t = 2$: magnetization vectors (left) and their magnitude on $y = 0.1$ (right).

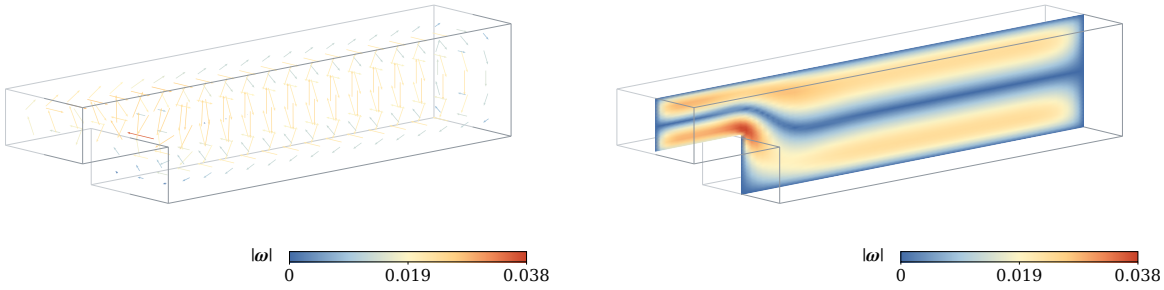


Figure 15: Backward-facing-step flow, third order in time and space, $K = 8$, $t = 2$: particle angular velocity vectors (left) and their magnitude on $y = 0.1$ (right).

A Absolute errors for the common manufactured-solution test

The following tables give the absolute errors for exactly the same seventeen cases as Section 6.1, with $T = 2$ and $\Delta t = 1/(2K)$. The six nonpressure fields use the terminal full norms; $\widetilde{p}_{\mathrm{stg}}$ uses the last-step stage-weighted L^2 norm defined there. The rates below are computed from the unrounded absolute errors, and hence may differ from the relative pressure rates. These tables reuse the same accepted numerical outputs; no additional computation is involved.

Table 5: Absolute errors and observed rates for the first-order scheme under joint space–time refinement.

(a) Velocity, stage pressure, angular velocity, and magnetization

K	$\boldsymbol{u}$: H^1		$\widetilde{p}_{\mathrm{stg}}$: L^2		$\boldsymbol{\omega}$: H^1		$\boldsymbol{m}$: $H(\mathrm{div})$	
	Error	Rate	Error	Rate	Error	Rate	Error	Rate
2	2.5008×10^{0}	—	9.7938×10^{0}	—	1.3043×10^{0}	—	7.3963×10^{-1}	—
4	1.3565×10^{0}	0.883	5.2188×10^{0}	0.908	7.0499×10^{-1}	0.888	3.9024×10^{-1}	0.922
8	6.4417×10^{-1}	1.074	1.7394×10^{0}	1.585	3.5944×10^{-1}	0.972	1.9752×10^{-1}	0.982
16	3.1038×10^{-1}	1.053	5.1197×10^{-1}	1.764	1.8059×10^{-1}	0.993	9.9085×10^{-2}	0.995
32	1.5263×10^{-1}	1.024	1.5141×10^{-1}	1.758	9.0407×10^{-2}	0.998	4.9593×10^{-2}	0.999

(b) Magnetic field, auxiliary curl, and scalar potential

K	$\boldsymbol{H}$: $H(\mathrm{curl})$		$\boldsymbol{k}$: L^2		φ: H^1	
	Error	Rate	Error	Rate	Error	Rate
2	1.6024×10^{0}	—	1.9637×10^{-1}	—	1.6325×10^{0}	—
4	8.7922×10^{-1}	0.866	1.0372×10^{-1}	0.921	8.8296×10^{-1}	0.887
8	4.4883×10^{-1}	0.970	5.0407×10^{-2}	1.041	4.4929×10^{-1}	0.975
16	2.2552×10^{-1}	0.993	2.4692×10^{-2}	1.030	2.2557×10^{-1}	0.994
32	1.1289×10^{-1}	0.998	1.2249×10^{-2}	1.011	1.1290×10^{-1}	0.999

Table 6: Absolute errors and observed rates for the second-order scheme under joint space–time refinement.

(a) Velocity, stage pressure, angular velocity, and magnetization

K	$\boldsymbol{u}$: H^1		$\widetilde{p}_{\mathrm{stg}}$: L^2		$\boldsymbol{\omega}$: H^1		$\boldsymbol{m}$: $H(\mathrm{div})$	
	Error	Rate	Error	Rate	Error	Rate	Error	Rate
2	6.6032×10^{-1}	—	8.8930×10^{-1}	—	3.0406×10^{-1}	—	1.5156×10^{-1}	—
4	1.7286×10^{-1}	1.934	1.6852×10^{-1}	2.400	8.0718×10^{-2}	1.913	3.9277×10^{-2}	1.948
8	4.3380×10^{-2}	1.994	3.4278×10^{-2}	2.298	2.0444×10^{-2}	1.981	9.8757×10^{-3}	1.992
16	1.0849×10^{-2}	1.999	8.1060×10^{-3}	2.080	5.1250×10^{-3}	1.996	2.4715×10^{-3}	1.998
32	2.7124×10^{-3}	2.000	2.0061×10^{-3}	2.015	1.2820×10^{-3}	1.999	6.1800×10^{-4}	2.000

(b) Magnetic field, auxiliary curl, and scalar potential

K	$\boldsymbol{H}$: $H(\mathrm{curl})$		$\boldsymbol{k}$: L^2		φ: H^1	
	Error	Rate	Error	Rate	Error	Rate
2	3.7683×10^{-1}	—	6.1889×10^{-2}	—	3.7797×10^{-1}	—
4	1.0056×10^{-1}	1.906	1.8154×10^{-2}	1.769	1.0063×10^{-1}	1.909
8	2.5577×10^{-2}	1.975	4.8169×10^{-3}	1.914	2.5582×10^{-2}	1.976
16	6.4236×10^{-3}	1.993	1.2378×10^{-3}	1.960	6.4239×10^{-3}	1.994
32	1.6078×10^{-3}	1.998	3.1330×10^{-4}	1.982	1.6078×10^{-3}	1.998

Table 7: Absolute errors and observed rates for the third-order scheme under joint space–time refinement.

(a) Velocity, stage pressure, angular velocity, and magnetization

K	$\boldsymbol{u}$: H^1		$\widetilde{p}_{\mathrm{stg}}$: L^2		$\boldsymbol{\omega}$: H^1		$\boldsymbol{m}$: $H(\mathrm{div})$	
	Error	Rate	Error	Rate	Error	Rate	Error	Rate
2	1.3515×10^{-1}	—	2.2456×10^{-1}	—	3.4212×10^{-2}	—	1.6509×10^{-2}	—
4	1.6877×10^{-2}	3.001	1.6686×10^{-2}	3.750	4.2421×10^{-3}	3.012	2.0774×10^{-3}	2.990
8	2.1487×10^{-3}	2.974	1.6515×10^{-3}	3.337	5.2312×10^{-4}	3.020	2.5949×10^{-4}	3.001
16	2.7338×10^{-4}	2.974	1.9746×10^{-4}	3.064	6.4765×10^{-5}	3.014	3.2407×10^{-5}	3.001

(b) Magnetic field, auxiliary curl, and scalar potential

K	$\boldsymbol{H}$: $H(\mathrm{curl})$		$\boldsymbol{k}$: L^2		φ: H^1	
	Error	Rate	Error	Rate	Error	Rate
2	4.1230×10^{-2}	—	1.0325×10^{-2}	—	4.1291×10^{-2}	—
4	5.1029×10^{-3}	3.014	1.4523×10^{-3}	2.830	5.1048×10^{-3}	3.016
8	6.2800×10^{-4}	3.022	1.8686×10^{-4}	2.958	6.2805×10^{-4}	3.023
16	7.7642×10^{-5}	3.016	2.3515×10^{-5}	2.990	7.7643×10^{-5}	3.016

Table 8: Absolute errors and observed rates for the fourth-order scheme under joint space–time refinement.

(a) Velocity, stage pressure, angular velocity, and magnetization

K	$\boldsymbol{u}$: H^1		$\widetilde{p}_{\mathrm{stg}}$: L^2		$\boldsymbol{\omega}$: H^1		$\boldsymbol{m}$: $H(\mathrm{div})$	
	Error	Rate	Error	Rate	Error	Rate	Error	Rate
2	1.5102×10^{-2}	—	1.6265×10^{-2}	—	1.9039×10^{-3}	—	9.7667×10^{-4}	—
4	9.1628×10^{-4}	4.043	6.7622×10^{-4}	4.588	1.1307×10^{-4}	4.074	5.9328×10^{-5}	4.041
8	5.6472×10^{-5}	4.020	2.9785×10^{-5}	4.505	6.9122×10^{-6}	4.032	3.6383×10^{-6}	4.027

(b) Magnetic field, auxiliary curl, and scalar potential

K	$\boldsymbol{H}$: $H(\mathrm{curl})$		$\boldsymbol{k}$: L^2		φ: H^1	
	Error	Rate	Error	Rate	Error	Rate
2	2.2772×10^{-3}	—	1.6977×10^{-3}	—	2.2789×10^{-3}	—
4	1.3692×10^{-4}	4.056	1.1892×10^{-4}	3.835	1.3695×10^{-4}	4.057
8	8.3748×10^{-6}	4.031	7.7699×10^{-6}	3.936	8.3752×10^{-6}	4.031

Declaration of Generative AI and AI-Assisted Technologies

The initial draft of this manuscript, including the mathematical model, the abstract formulation, and the main mathematical derivations, was written and developed independently by the authors. During the revision process, generative AI was used only to assist with improving the English expression and the logical organization of the presentation, with the aim of making the manuscript clearer and easier to read.

Generative AI was also used as an auxiliary tool in the construction of the higher-order time discretizations. In particular, within the abstract framework developed by the authors, it assisted in identifying candidate Runge–Kutta coefficients for the third- and fourth-order methods that satisfy the required stage-order, quadrature, and energy-stability conditions. All AI-assisted suggestions, including the coefficients, formulas, and related arguments, were independently checked and verified by the authors. The authors take full responsibility for the mathematical correctness, originality, and final content of the manuscript.

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